Macro and Micro Dynamics of Productivity: From Devilish Details to Insights

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CES 17-41 **May, 2017**

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Abstract

Researchers have been using a variety of methods to estimate productivity at the firm level. Absent data on prices and quantities, these methods yield what have become known as revenue productivity measures. How these measures are related to physical productivity depends on the assumptions about the environment in which establishments operate. It is perhaps less recognized that the differences across estimation methods have important consequences for interpretation. One such difference concerns revenue function estimates: while cost-share-based coefficients are, in principle, equivalent to factor elasticities, regression-based estimates equal factor elasticities only under strict assumptions about product markets. This implies that revenue residuals are conceptually different under these two broad approaches. Using plant-level manufacturing data for the U.S., we look at the empirical relevance of such distinctions in the context of key stylized facts of the productivity literature. First, we find non-trivial differences in estimated elasticities and returns to scale. The variation in elasticities affects numerical results on dispersion, yet all methods imply large productivity differences across establishments. More productive plants are shown to be more likely to grow and survive by all reviewed methods, although differences remain in the quantitative marginal effects of productivity. Reallocation is found to be comparable and productivity enhancing by all methods considered, but within-plant growth seems to be more sensitive. We find evidence that imputation and imposing homogeneous elasticities negatively affect within-industry dispersion. In addition, imputation results in some attenuation in growth and exit coefficients but does not invalidate qualitative conclusions.

* We would like to thank conference participants at the 2013 Comparative Analysis of Enterprise Data in Atlanta, 2014 Research Data Center Annual Conference and the 2015 NBER CRIW workshop for valuable comments. We are grateful to Kirk White for useful discussions and for making his code available to us. Any remaining errors are our own. Any conclusions expressed herein are those of the authors and do not necessarily represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. † Corresponding author: zoltan.wolf@census.gov.

1 Introduction

A ubiquitous and influential finding in the empirical literature on firm dynamics is that there is large dispersion in measured productivity across establishments within narrowly defined industries. This finding has generated much analysis of the causes and consequences of such dispersion. Explanations of possible causes include curvature in the profit function that prevents the most productive firm from taking over an industry, frictions in adjustment of factors and the entry and exit of plants, and distortions that drive wedges in the forces pushing towards the equalization of marginal products across plants. In terms of consequences, there is a burgeoning literature on the connection between reallocation dynamics, growth and productivity. Many papers have found that more productive plants are more likely to grow and less likely to exit, implying that the reallocation of inputs observed across plants is productivity enhancing. In like fashion, there is increased attention to reasons why these reallocation dynamics may vary over the business cycle and across countries and in turn how these account for differences in economic performance across time and countries. In an important related area of inquiry, a recent theoretical and empirical literature hypothesizes that gains from opening markets to trade are due to the improved allocation of resources induced by trade.[1]

While there is considerable consensus that accounting for the dispersion of productivity and its connection to reallocation are important for understanding differences in economic performance, there is not a consensus about the basics of estimating plant-level productivity. Given that typical micro datasets contain information about revenues and input expenditures but not quantities, the majority of results are based on what have become known as revenue productivity measures. How these measures are related to physical productivity depends on the assumptions about the environment in which establishments operate.

A variety of methods are available to researchers to estimate revenue productivity. One approach, actively used by the statistical agencies in official aggregate and industry-level productivity statistics, is based on the cost-shares of input expenditures. We will call the productivity measure implied by this approach tfpr^{cs}. If plants are cost-minimizing and the assumption of constant returns to scale holds, then cost-shares are valid estimates of factor elasticities and the implied revenue productivity measure will be a product of output prices and physical productivity. Other approaches estimate elasticities using regression techniques. We will denote the implied revenue productivity measure by tfpr^{rr}. Econometric issues aside[2], the most important property that distinguishes cost-share-based methods from regression-based techniques is the fact that absent data on quantities or further assumption about demand, the latter yields revenue elasticities, while the former in principle yields factor elasticities even when using data on revenues and expenditures.[3] This implies these revenue productivity measures are concep-

[1] See the survey in Syverson (2011) for relevant cites to the findings in the literature and the theoretical and empirical literature that has developed in light of the large dispersion in productivity and its relationship to reallocation, growth and aggregate productivity.

[2] Since ordinary least squares is problematic given the endogeneity of factor inputs, alternative estimation methods based on instrumental variables techniques have been developed to address this endogeneity. While these methods are attractive in principle, they are not commonly used given the lack of plausible instruments on a wide scale basis to cover all industries over all time periods. Instead, methods have been developed that seek to deal with the endogeneity issues by using proxies for the productivity residuals. Numerous alternative proxy estimation approaches have been developed with a robust debate about the respective merits of the each approach. Different approaches make alternative assumptions about the timing of input decisions with respect to the realization of shocks and the relative importance of addressing the impact of selection on estimation.

[3] Although earlier papers recognized the difference between the two, researchers have not focused on such distinctions. We believe this is due to the fact that, numerical differences notwithstanding, results under

tually different. The two will be equal only in the case when output is homogeneous and/or there is no price heterogeneity. Under these circumstances, revenue productivity differences across plants can be interpreted as differences in technical efficiency.[4] Decomposing revenue elasticities into factor elasticities and demand elasticities requires more assumptions about demand and alternative estimation techniques. Exploring the role of demand has become an active area of research in recent years,[5] and existing evidence suggests revenue residuals reflect both technical efficiency and demand side factors and the estimated revenue elasticities reflect both the technology and the demand structure. Exactly how these demand side factors impact the interpretation of exercises resulting from a specific estimation method will depend on the assumed demand structure.

In this paper, our objective is to assess the empirical relevance of such distinctions. First, we compare cost-share-based elasticities to revenue elasticities. To interpret these we follow the approach described in Klette and Griliches (1996) to derive factor elasticities after jointly estimating revenue elasticities and demand elasticities. In addition, we find sizeable differences among regression based methods themselves, particularly in terms of capital elasticities and therefore estimated returns to scale. Second, our results show that cost-share-based measures and regression-based measures have different implications for the productivity rank of plants within industries, confirming earlier findings on the differences between these methods. Third, in spite of these differences, the basic facts about productivity dispersion remain largely robust in the sense that all methods yield considerable within-industry dispersion of productivity. However, there are quantitative differences and since the precise estimate of productivity dispersion matters for some analyses this variation is potentially quite important. Fourth, our reduced form regressions suggest both approaches have generally similar implications about the relationship between growth, survival and productivity at the establishment level. Numerical differences aside, more productive establishments are found to be more likely to grow and survive.

Finally, we assess whether the distinction between the two broad approaches matters for the conclusions of the structural decomposition literature on aggregate productivity growth (APG). We show that the difference between revenue and factor elasticities can be important and assess whether this distinction is empirically relevant by implementing the decomposition proposed by Petrin and Levinsohn (2012). We find that all estimation methods yield a substantial contribution of reallocation to APG. However, there is variation across methods in terms of the quantitative significance of within-plant productivity growth, fixed costs and frictions.

The generally similar conclusions we obtain in this paper have an important implication for a popular area in productivity research. Hsieh and Klenow (2009) show that, under specific assumptions about demand and production technology, the dispersion in cost-share-based productivity measures reflects the effect of distortions. In contrast, Foster, Grim, Haltiwanger, and Wolf (2016b) (hereafter FGHW) show that dispersion in regression-based productivity measures are a function of physical productivity and demand shocks. These theoretical findings could in principle be justified by the conceptual differences mentioned above. On the other hand, our empirical findings in this paper indicate that these productivity indicators are positively corre-

alternative estimation methods show some degree of similarity.

[4]If prices are heterogenous reflecting exogenous quality differences in outputs and inputs, then variation in revenue productivity reflects differences in technical efficiency adjusted for quality differences.

[5]See, e.g., Foster, Haltiwanger, and Syverson (2008) and De Loecker (2011) on the difference between revenue productivity and physical productivity. FGHW (hereafter FGHW) further investigate the implications of modeling demand in the context of the recent literature on misallocation.

lated and that their implied dispersion is comparable. This is an interesting finding because if the assumptions in Hsieh and Klenow (2009) hold, they imply distortions are positively correlated with demand and technical efficiency shocks. One could argue for correlated distortions but this explanation raises the question why more distorted firms are more likely to be hit by positive demand shocks and technical efficiency shocks. An alternative interpretation attributes dispersion to the effect of adjustment frictions. In the presence of such frictions, plants do not adjust production factors instantaneously and therefore high-productivity plants will exhibit high revenue productivity estimates.

We think the robustness of the core findings in the literature has not been settled. In his survey, Syverson (2011) discusses this debate and observes that many papers in the literature explore the robustness of their findings to alternative estimation methods. Our reading, like his, is that many papers report that results are reasonably robust to alternative estimation methods. But most papers focus on a specific question often for a narrow set of industries. It is less clear to us how robust core findings are to these issues. The literature offers some guidelines, based on Monte Carlo evidence[6], as to which method is optimal in the presence of certain types of measurement or specification error. Important examples include heterogeneity in input prices, technologies or measurement error in output, primary inputs, or some combination of these. These experiments tell us which estimation method is optimal conditional on isolated factors. While such evidence is valuable, in practice there may be a variety of sources of error. An equally important point is that existing Monte Carlo studies do not take into account the wide disparities in sample size that are available for empirical estimation. A limitation of our approach is that we cannot and do not take a stand on the optimal method. Instead, our objective is to identify the nature of the sensitivity of key results to the distinction described above.

The paper is organized as follows. We discuss our methodology and data in Sections 2 and 3. Section 4 describes the effect of estimation methods on the distribution of elasticity estimates. Section 5 describes the implications of the differences in elasticity distributions on productivity dispersion, plant growth and survival, and aggregate productivity growth decompositions. Section 6 concludes.

2 Methodology

2.1 Productivity measures

A useful starting point is a conceptual measure of revenue per composite input under Cobb-Douglas technology, that allows the decomposition of demand side and supply side factors. Following the notation in FGHW, this measure can be written in logs as

$$\text{tfpr}_i = p_i + q_i - \sum_j \alpha_j x_{ij} = p_i + \text{tfpq}_i, \tag{1}$$

where i indexes plants, p, q and x denote output prices, quantities and inputs, respectively. Equation (1) makes explicit that tfpr_i confounds the effect of output prices and physical productivity, denoted by tfpq_i. Given that typical micro datasets contain information about revenues and input expenditures but not quantities, this property has the important implication that

[6]Van Biesebroeck (2007) explores the behavior of a number of estimators in the presence of measurement error and parameter heterogeneity. Martin (2008) investigates a case when variable inputs are measured with error.

the majority of results in the empirical productivity literature is based on what have become known as revenue productivity measures.

One typical measure is equivalent to tfpr_i if one is willing to assume that returns to scale is constant. The equality holds because under this assumption, cost shares equal their respective factor elasticities, denoted by α_j in equation (1).[7] Formally,

$$\text{tfpr}_i^{cs} = p_i + q_i - \sum_j \alpha_j^{cs} x_{ij} = \text{tfpr}_i. \tag{2}$$

Other estimation methods rely on regression techniques to obtain revenue productivity. We denote these measures by tfpr_i^{rr} in order to distinguish them from tfpr_i^{cs}. Absent data on prices and/or quantities, these methods yield revenue function estimates or elasticities, which are a function of factor elasticities and demand parameters, in general. As a result, tfpr_i^{rr} depends on both physical productivity and demand shocks and is not equal to either tfpr_i or its estimate, tfpr_i^{cs}:

$$\text{tfpr}_i^{rr} = p_i + q_i - \sum_j \beta_j x_{ij} \neq \begin{cases} \text{tfpr}_i \\ \text{tfpr}_i^{cs} \end{cases} \tag{3}$$

To highlight the importance of the difference relative to tfpr_i^{cs} and tfpr_i, we denoted *revenue elasticities* by β_j in equation (3). It can be shown that, under the assumption of isoelastic demand, factor elasticities are equal to the ratio of revenue elasticities and a parameter that determines the price elasticity of output: $\alpha_j = \beta_j/\rho$. We will discuss this relationship in more detail in section 4 and highlight the role of such distinctions in subsequent analyses where appropriate.

2.2 Estimation methods

There are various estimation methods available to researchers; table 1 summarizes the procedures we use throughout the paper. We discuss the strengths and weaknesses of the most popular methods from a practical point of view.

Table 1: Estimation methods.

Method	Description	Proxy	Estimator	Productivity measure
GA	Foster, Haltiwanger, and Krizan (2001)		Cost-shares	tfpr^{cs}
OLS	Ordinary Least Squares		LS	tfpr^{rr}
OP	Olley and Pakes (1996)	Investment	NLS	tfpr^{rr}
LPGR	Levinsohn and Petrin (2003)	Materials	GMM	tfpr^{rr}
WLPE	Wooldridge (2009)	Energy	Efficient GMM	tfpr^{rr}
WLPM	Wooldridge (2009)	Materials	Efficient GMM	tfpr^{rr}

All productivity measures in the table are estimated using an output-based production function.

Cost-share-based or growth accounting methods (GA) exploit first order conditions from the firm's cost-minimization problem. There are several advantages to this procedure. For example, this is the only estimator that - conditional on an assumption about returns to scale

[7] It can be shown that given output and input prices and a measure of returns to scale, the shares of input expenditures in the plant's total cost are equal to the corresponding factor elasticities.

- provides direct estimates of factor elasticities without data on prices and quantities. This is a useful property because it implies cost-share-based elasticities are robust to alternative demand structures. In addition, using the share of input expenditures in total costs rather than in value added has the advantage that the assumption of perfectly competitive markets is not required. In addition, it allows the exact shape of technology to vary even at the plant-level.[8] A potential caveat is that a known estimate of, or an assumption about, returns to scale is necessary. In addition, elasticity estimates may be biased if the first order conditions are violated. This property is relevant because it is unlikely that first order conditions hold for every plant at all points in time. Therefore, common factor elasticities across plants in the same industry and/or over time are frequently imposed (see Syverson (2011)). We explore the implications of these assumptions further in appendix A.5.3 and note that all alternative estimation methods discussed in this paper require such restrictions.

Ordinary least squares (OLS) estimates of elasticities are inconsistent because unobserved productivity shocks affect the decision problem of plants, see Marschak and Andrews (1944). Additional biases may result if prices are unobserved and correlated with factor inputs, see Klette and Griliches (1996).

The remaining methods we examine belong to a class often referred to as proxy methods because they use firm-level indicators to control for the effect of unobserved productivity during estimation. The original logic, developed in Olley and Pakes (1996) (OP hereafter), is based on assuming a monotonous and increasing relationship between the proxy and productivity. On condition that productivity is the only unobserved state variable, plant-level variation in the proxy can be interpreted as a reflection of the variation in productivity shocks. This conclusion together with timing assumptions about the plant's decisions, can be used to recover all the elasticities.[9] OP propose investment as a proxy. There is ample evidence that plant-level investment is lumpy (see for example Cooper and Haltiwanger (2006)), and lumpiness means bursts of investment activity are followed by inactive periods where observed net investment is zero rendering these observations uniformative for OP. To eliminate the potential efficiency loss, Levinsohn and Petrin (2003) (LP hereafter) advocate using intermediate input expenditures or energy costs.[10] LP argue that if adjusting intermediate inputs is less costly, then they are likely to be more responsive to productivity shocks. This is especially relevant in the presence of non-convexities in capital adjustment.

The identifying assumptions regarding the timing of plants' input decisions have been criticized by Ackerberg, Caves, and Frazer (2015) (ACF hereafter). ACF argue that the optimal labor allocation is also a deterministic function of productivity and therefore the labor elasticity is not identified in the first step. They approach the identification problem by applying a two step procedure that does not try to identify any of the elasticities in the first stage. Wooldridge

[8]Further, evidence in Van Biesebroeck (2007) suggests that this method is accurate if the data are not subject to much measurement error.

[9]Proxy methods are multi-step procedures, where the first-step-estimates of variable input elasticities are determined by OLS and the coefficients of quasi-fixed inputs such as capital are separately identified using moment conditions formed by estimated productivity innovations and lagged capital values. These methods rely on polynomial approximations at two points of the estimation algorithm. First, a polynomial of the state variables and the proxy is included to approximate unobserved productivity. Second, the algorithm exploits a Markovian assumption about the plant-level productivity process in order to extract productivity innovations. It is important to note that while polynomial series approximations are flexible, higher order terms may exacerbate measurement error likely present in microdata.

[10]LP highlight that firms almost always report positive use of these variables in their data implying truncation due to zero proxy values is less severe.

(2009) proposed to circumvent the identification problem by estimating all the coefficients in a single GMM step and using earlier outcomes of both capital and variable inputs as instrumental variables. His approach is advantageous because it is robust to the ACF critique and because the efficiency loss that arises from two-step estimation is eliminated.

In closing this section we note that estimating factor elasticities under endogenous plant-level prices requires additional assumptions about demand, see Klette and Griliches (1996) or De Loecker (2011) as examples. We will see in section 4 that such considerations are empirically relevant for inferences about returns to scale.

3 Data

3.1 Source data

Our industry-level data, including deflators, capital rental prices and depreciation rates, are taken from the NBER-CES Manufacturing database[11], the Bureau of Labor Statistics and the Bureau of Economic Analysis. We use establishment-level information from the Annual Survey of Manufactures (ASM), Census of Manufactures (CM) and the Longitudinal Business Database (LBD).

The CM collects data in years ending in '2' and '7' for roughly 180,000 - 240,000 plants. Establishments with less the five employees are not sent forms. Payroll and employment data for these very small plants are imputed using administrative records.[12] The ASM surveys 50,000-70,000 establishments in non-Census years and is part of the CM in Census years. It is a rotating panel re-defined two years after the latest Census. The LBD contains the universe of non-agricultural business establishments with paid employees and is based on both survey information and administrative records. Appendix A in Foster, Grim, and Haltiwanger (2016a) (FGH, hereafter) describes these data in more detail. Our initial dataset includes approximately 3.5 million plant-year observations between 1972-2010.

We use the ASM and CM to construct plant-level measures of inputs and output. Output is measured as a deflated value of total value of shipments, corrected for the change in finished goods and work-in-process inventories. Labor input, total hours worked, is constructed as the product of production worker hours and the ratio of the total wage bill to production worker wages. Our intermediate input variable is given by the the sum of three items: cost of parts, contracted work and goods resold. The energy input consists of deflated electricity and fuel costs. We create establishment-level capital stock measures using a version of the Perpetual Inventory Method, which calculates current capital as a sum of the depreciated stock and current investment. We set plants' initial capital stock to a deflated book value taken from the ASM and CM. More details on the construction of input and output measures can be found in appendix B of FGH. The LBD serves two purposes in our analysis. First, high-quality longitudinal identifiers help us determine the accurate time of establishments' exit which is a necessary indicator to estimate the relationship between productivity, growth and exit. Second, the LBD acts as a universe file; we use employment and establishment age data from the LBD to construct inverse propensity score weights that control for non-randomness in our sample.[13]

[11]The NBER-CES Manufacturing Industry database is available at http://www.nber.org/nberces. An earlier version is documented in Bartelsman and Gray (1996).

[12]We drop administrative records cases.

[13]Employment data is useful to determine the probability of size-based selection into the ASM and CM. Establishment age is an important determinant of the probability that the productivity level of an establishment is calculated from imputed data.

3.2 Analysis samples

The analyses in this paper exploit three different samples. For questions about the distributions of elasticities, productivity dispersion and growth and survival, we look simultaneously at two samples, which we refer to as the 10 and 50 largest industry samples. These samples need to fulfill two potentially contradicting requirements. First, the number of plant-year observations within each industry should be large enough so that elasticities can be estimated by all reviewed methods. Second, industries should be defined narrowly enough so that we can plausibly assume elasticities are constant across establishments.

Changes in industry classification systems over time make defining these samples more complicated than simply choosing the 10 and 50 industries with the largest number of plant-year observations. Since we estimate elasticities on an industry-by-industry basis, changes in the classification system entail spurious breaks in plant-level time series and a drop in sample size. In the first part of the analysis we address these issues by selecting sets of 4-digit SIC industries which were not affected by changes or which were mapped one-to-one into another industry. There are 292 such industries of which we selected the first 10 and 50 based on the number of observations.

We create a third dataset to test whether the implications of the decomposition of aggregate productivity growth described in Petrin, White, and Reiter (2011) (PWR hereafter) are sensitive to the way productivity is estimated. Since we are attempting in part to replicate the results in PWR, we create a roughly comparable dataset. PWR's data spans the period between 1976-1996 so the 1987 change in SIC classification is relevant. To correct for these breaks, we follow the first step of PWR's procedure and assign the SIC code to any establishment observed between 1987-1996. However, we deviate from their approach for cases only observed prior to 1987. If a plant is not assigned an industry code in the previous step, we apply a random assignment procedure based on the share of shipments mapped from the 1972 to 1987 SIC industry code. More details about the assignment procedure can be found in appendix A.2.

4 Elasticity distributions

We start by discussing differences in the distribution of capital elasticities. Next, we check whether the elasticity-rank of industries varies with estimator choice. We conclude the section by looking at the implications of estimator choice on returns to scale.

Figure 1 plots the density estimates of capital elasticities in the 50 most populous industries and table 2 shows basic descriptive statistics. There are non-trivial differences in the mean, dispersion and general shape of the distributions. Most notably, GA-based estimates tend to be generally smaller than the rest. This is to be expected because GA-estimates are equivalent to factor elasticities under the assumption of cost minimizing behavior and constant returns to scale. In contrast, regression-based estimates yield revenue elasticities, which capture both factor elasticities and demand side factors.

We now turn our attention to the revenue elasticity distributions and focus on indications of possible biases in the discussion. Unfortunately, the differences in estimated distributions alone do not tell much about the direction, much less the magnitude, of any bias. However, in light of what we know about the way these methods address endogeneity, the differences may give us clues as to whether or not they correct it in the right direction. The direction of the bias in OLS-estimates is determined by several factors. First, since input demand functions are increasing in productivity, OLS estimates are biased upward. If this is important in our data and proxy methods correct for it, then we should see proxy-based distributions to the left of OLS.

Figure 1 suggests that only LPGR is likely to yield lower $\hat{\beta}_k$. However, the direction of the bias depends on additional factors. For example, Levinsohn and Petrin (2003) show that positive correlation between capital and labor may cause $\hat{\beta}_k$ to be biased downward, implying that the elasticity distribution may emerge to the left of OLS not because LP corrects an upward bias but because it includes a downward bias. A further complicating factor is selection. OP argue that since plants' profit and value functions are increasing in capital, larger establishments anticipate larger future returns and therefore can operate at lower current productivity levels, which also entails a negative bias in OLS. If OP corrects for such selection-induced negative bias, and this effect is important in our data, then the OP-based $\hat{\beta}_k$ distribution should be to the right of OLS.

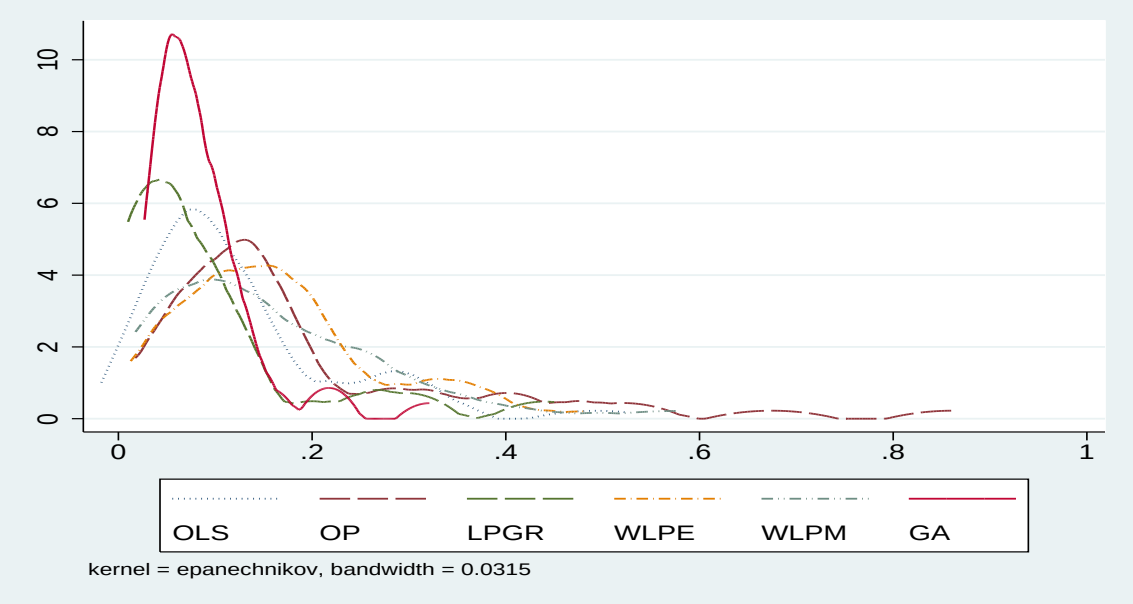


Figure 1: Cross-industry distributions of $\hat{\beta}_k$.
The sample is 50 largest industries, estimators are described in table 1.

Table 2: Descriptive statistics of the between-industry distribution of $\hat{\beta}_k$.

	Mean	Median	IQR	Mean	Median	IQR
	50 largest industries			10 largest industries		
OLS	.13	.10	.10	.14	.12	.15
OP	.19	.14	.10	.21	.15	.27
LPGR	.09	.06	.09	.11	.06	.09
WLPE	.16	.15	.11	.17	.14	.07
WLPM	.16	.12	.14	.15	.09	.14
GA	.08	.07	.05	.09	.10	.06

See notes to table 1 for method definitions. 10 and 50 largest industries: most populous 4-digit industries industries which were mapped 1-to-1 between classification systems.

We find that OP tends to result in higher $\hat{\beta}_k$ than OLS suggesting that controlling for selection-induced bias may be important. As for other proxy methods, WLPE and WLPM are more likely to yield extreme $\hat{\beta}_k$ even though the typical elasticities under these methods are similar to those under OP. These results tell us that, in addition to proxy choice and addressing

selection, the structure of the estimator - non-parametric two-step estimator versus efficient GMM - may also have important consequences for capital elasticities. Similar conclusions hold for the second moments of $\hat{\beta}_k$. The interquartile range measures, see the last column of table 2, suggest that differences in the dispersion of these distributions are also non-trivial. For example, the means of GA and LPGR are very close (.08 and .09 among the 50 largest industries) but the difference in dispersion is almost twofold (.05 and .09). Finally, the elasticities of variable inputs show stronger clustering, especially $\hat{\beta}_l$ (figure 2). This is partly explained by the fact that proxy methods estimate β_l in an OLS step. The main conclusions about $\hat{\beta}_e$ and $\hat{\beta}_m$ are the same, (figure A3): there are numerical differences in these distributions but they look generally similar across estimation methods.

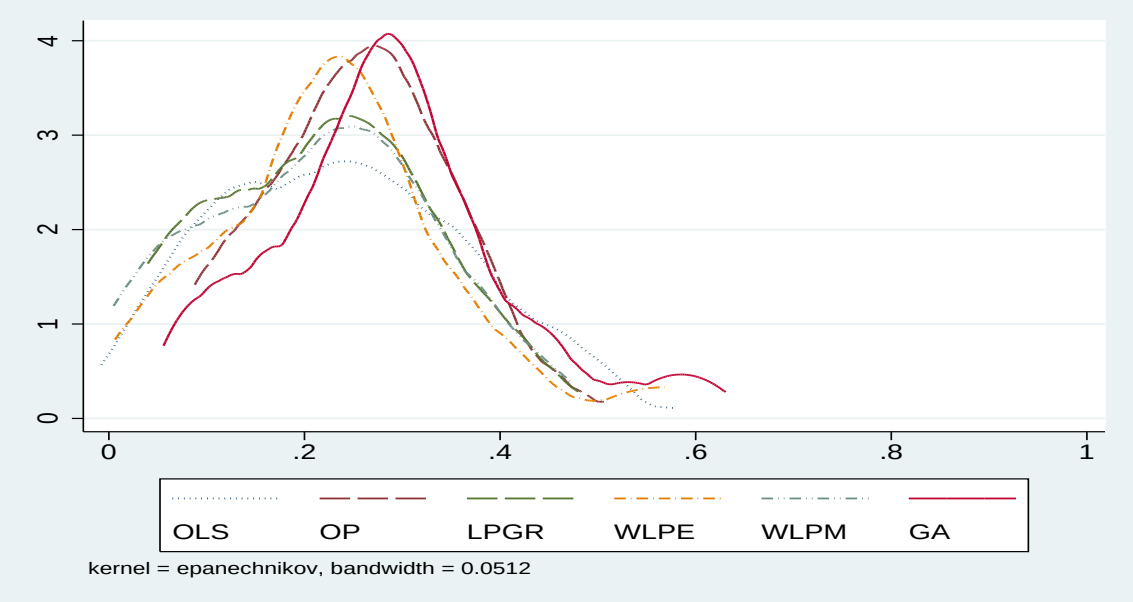


Figure 2: Cross-industry distributions of $\hat{\beta}_l$.
The sample is 50 largest industries, estimators are described in table 1.

Next, we explore whether the choice of estimator affects the ranking of industries by $\hat{\beta}_k/\hat{\beta}_l$. If the estimation method is not crucial then industries' rank should not vary across methods. Our results suggest the opposite: we find there is a positive probability that different estimators imply different industry rankings, see table A10 for details.[14]

In closing this section, we would like to highlight the different implications estimation methods have for returns to scale. GA-based elasticities are consistent with constant returns to scale, by construction. In contrast, regression based revenue function estimators do not impose constant returns to scale but do not provide direct returns to scale estimates either. However, if one is willing to accept the assumption that plants' face isoelastic product demand, joint estimation of revenue elasticities and demand characteristics can be used to make inference about factor elasticities and therefore returns to scale. We follow Klette and Griliches (1996), and jointly estimate revenue function coefficients (β_j) and a parameter that determines the price elasticity of product demand (ρ) by regressing plant-level revenues on plant-level inputs and an indicator

[14]We further investigate the empirical properties of elasticity distributions in Appendix A.3. Those findings highlight that some methods yield systematically more zero/negative elasticities than others. We find evidence that, in certain cases, this can be attributed to small sample size and/or proxy choice.

of industry-level demand. It can be shown that under isoelastic demand conditions, there is a one-to-one mapping between the coefficient of industry-level demand (β_q) and the demand parameter: β_q=1-ρ, implying that we can recover factor elasticities as $\hat{\alpha_j}=\hat{\beta}_j/\hat{\rho}$, and therefore returns to scale $\sum_j \hat{\alpha_j}$, see also FGHW and the web appendix therein for more details. Our results, based on OP's method,[15] indicate that the elasticity of firm revenues with respect to aggregate demand is a small positive number, implying a negative average price elasticity and greater-than-1 markup value, see table A11 for more details. Consequently, factor elasticities are larger than revenue elasticities in the average industry. However, the cross-industry distributions of these parameters suggest one should exercise caution when interpreting industry specific results: the implied price elasticity estimate varies so much that it may even be positive in certain industries which can be interpreted as a sign that demand parameters are poorly estimated in these industries. To further illustrate the variation in these parameters, we compared the densities of the sums of revenue and factor elasticities, see figure 3. The sum of estimated factor elasticities shows large dispersion reflecting the variation both in revenue elasticities and demand parameters suggesting that appropriate caution is needed when interpreting results, especially on an industry-by-industry basis.

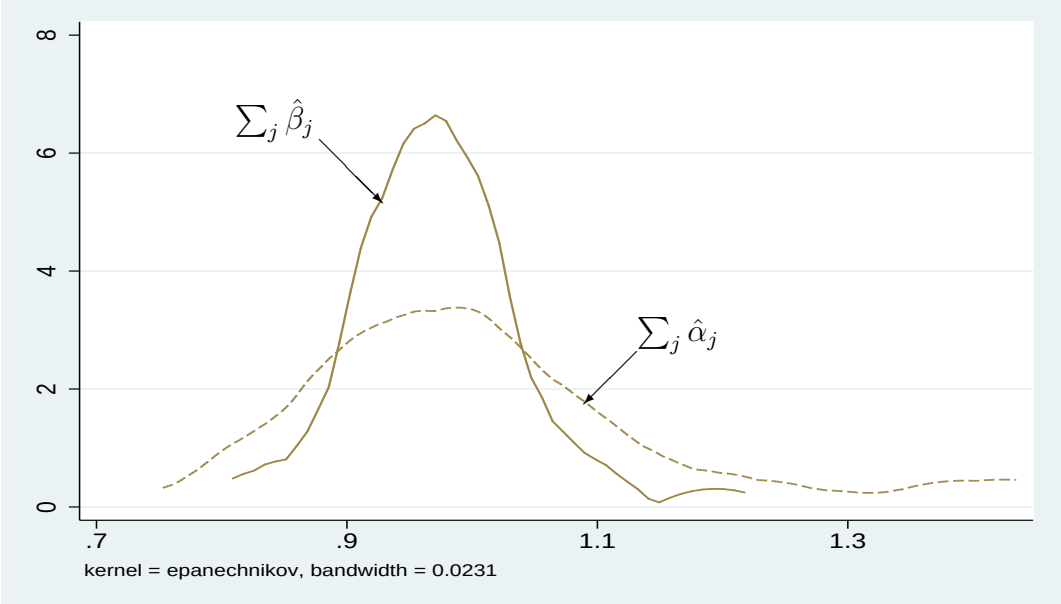


Figure 3: Cross-industry distributions of $\sum_j \hat{\beta}_j$ and $\sum_j \hat{\alpha}_j$

$\hat{\beta}_j$ and the demand parameter are jointly estimated using the method described in Olley-Pakes (1996).

5 Implications of the differences in elasticity distributions

In this section, we investigate whether the differences in elasticities matter for the core findings in the empirical productivity literature. Specifically, we examine: productivity dispersion; the relationship between productivity, growth and survival; and structural decompositions of aggregate productivity growth (APG).

[15] We chose OP because adding more state variables in the available Stata implementation is straightforward.

5.1 Productivity dispersion

Does the choice of estimator affect the second moment of the within-industry productivity distribution? The answer is yes and no. The interquartile range (IQR) and standard deviation, averaged over industries and time, suggest that differences in dispersion are nontrivial (table 3, more details in table A12). The average IQR of the log-productivity distribution varies between 0.24 and 0.4 across methods. These values imply that the plant at the 75th percentile in the average industry generates approximately 27-49% more revenue with the same amount of inputs than the plant at the 25th percentile. However, the differences across methods suggest one must exercise caution when using results for economic analysis where small changes in magnitudes may make a big difference.

The choice of estimation method has consequences also for the within-industry productivity rank of establishments. Table 4 shows that both the Pearson and Spearman rank correlations are substantially smaller than 1 implying that where plants sit in the productivity distribution is sensitive to choice of estimation method.[16] Such differences notwithstanding, the results confirm that productivity differences across establishments are invariably large no matter how we measure it.

Table 3: Descriptive statistics of productivity distributions.

	N (1000)	IQR	SD	N (1000)	IQR	SD
	50 largest industries			10 largest industries		
OLS	455	0.26	0.26	185	0.22	0.20
OP	380	0.32	0.38	152	0.31	0.39
LPGR	457	0.29	0.31	188	0.29	0.28
WLPE	457	0.34	0.36	187	0.31	0.28
WLPM	457	0.40	1.88	188	0.33	0.35
GA	433	0.24	0.22	177	0.23	0.21

See notes to tables 1 and 2 for the description of methods and samples. All statistics are based on deviations of plant-level log-productivity from industry- and time-specific means. All results shown were calculated using non-outlier observations only (pre-, post-estimation). A version of the table including pre-estimation outlier observations can be found in the appendix (table A12), results barely change.

Section 4 indicated that choosing one estimation method over another may have consequences for elasticity distributions. The results in this section so far suggest that different estimation methods imply invariably large dispersion, at least on average across industries and time. However, table 3 demonstrates there are numerical differences. Should we think of these as substantial? One way to answer this question is to test their equality by comparing confidence bands around average dispersion or comparing industry-specific results. Figure 4 shows simulated two-standard-error-wide confidence bands (narrow bars) around our dispersion estimates. If we interpret the narrow bars as approximately 95% confidence intervals, we can conclude that dispersion measures under proxy methods are not significantly different from each other. In other words, the sampling variation in revenue function estimates has similar effects proxy-based estimates.[17] The exception is GA, which generally yields lower dispersion.

Note that even if different methods yield similar average dispersion, industry-specific implications may differ. To highlight the implications of estimators in economic analyses where

[16] WLPM yields especially low correlations with other methods.

[17] WLPM is more likely to yield extreme elasticity estimates and therefore dispersion. More details can be found in table A16.

Table 4: Correlations among within-industry productivity distributions, 50 largest industries.

	OLS	OP	LPGR	WLPE	WLPM	GA
Pearson						
OLS	1					
OP	0.51	1				
LPGR	0.82	0.43	1			
WLPE	0.51	0.46	0.52	1		
WLPM	0.02	0.02	-0.15	0.15	1	
GA	0.79	0.46	0.68	0.51	0.09	1
Spearman						
OLS	1					
OP	0.68	1				
LPGR	0.87	0.61	1			
WLPE	0.59	0.61	0.56	1		
WLPM	0.35	0.36	0.26	0.49	1	
GA	0.81	0.63	0.70	0.60	0.43	1

Correlations reflect distributional differences discussed above: proxy methods show greater similarity, while WLPM seems more different. Rank correlations confirm. Including pre-estimation outliers has a minor effect on correlations (see table A13).

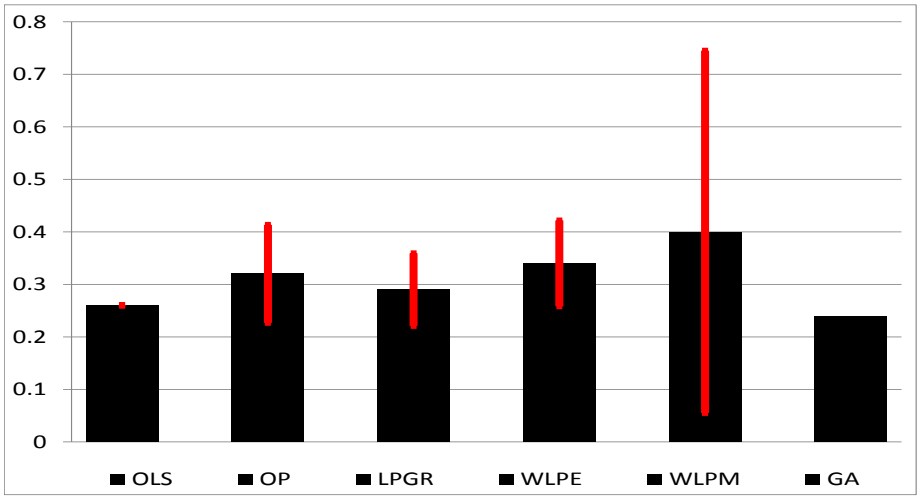
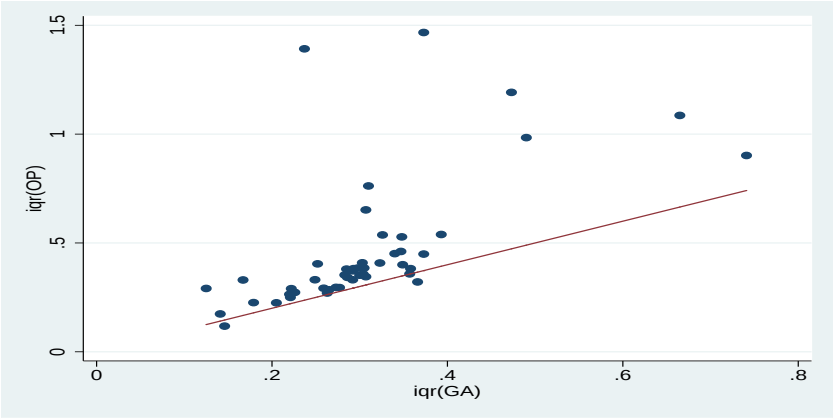
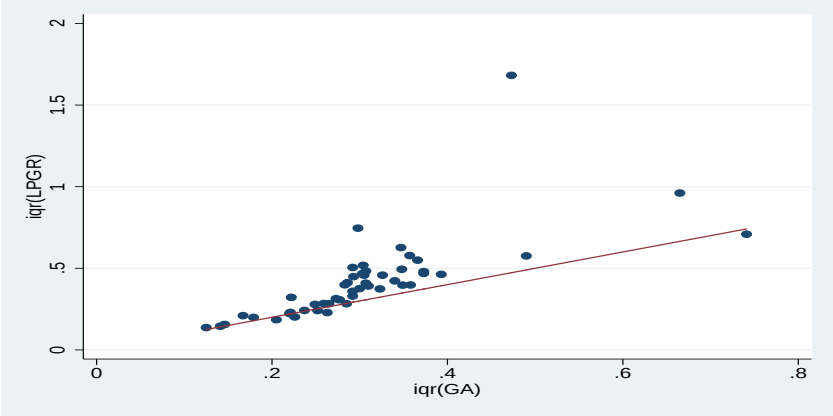


Figure 4: Bootstrapped standard errors of dispersion under selected methods.
The black bars are mean dispersion estimates from table 3, the red bars represent confidence bands based on bootstrapped standard errors of dispersion statistics, see also table A16. GA has no interval around the mean because coefficients are estimated using industry level data on input costs.

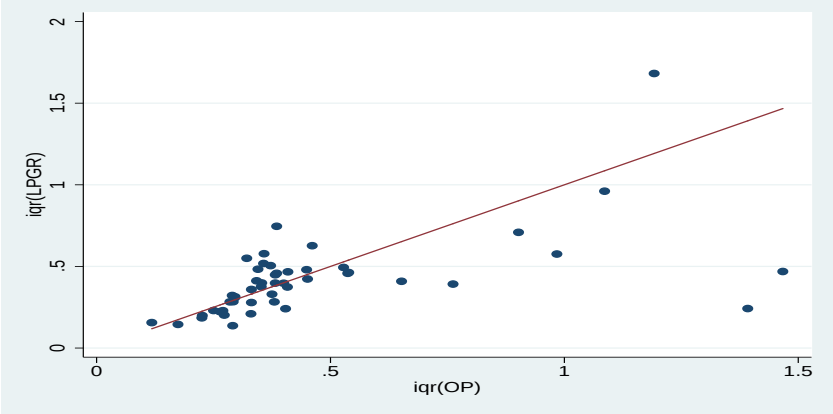
differences in magnitudes may matter a lot, we compare these measures across a selected set of estimators in figure 5. In panels (a)-(b), almost all points are located above the 45-degree line, which means that proxy methods (OP, LPGR) yield systematically larger dispersion than cost-share-based methods (GA). One explanation behind such divergence may be related to the conceptual differences mentioned earlier. On the other hand, comparing dispersion under LPGR and OP suggests that differences between two methods of the same class may also exist, see panel (c). These differences, smaller and less clear-cut than in the previous two panels, are still non-negligible. They can be attributed to technical aspects like the chosen econometric



(a) OP and GA



(b) LPGR and GA



(c) LPGR and OP

Figure 5: Industry-specific dispersion under selected methods.
Each point shows a pair of industry-specific dispersion measures and the line represents the 45-degree line in each chart.

procedure, and/or more substantive factors like the choice of proxy.

The main findings in this section have important implications for a popular area of inquiry in the productivity literature. Hsieh and Klenow (2009) show that, under isoelastic demand and constant returns to scale technology, the dispersion in cost-share-based productivity measures (or tfpr^{cs}) reflects the effect of distortions. In contrast, FGHW show that the dispersion in

regression-based productivity measures (or $tfpr^{rr}$) does not have this property and demonstrate that revenue function estimation yields productivity measures that are a function of physical productivity and demand shocks. Our results in this paper indicate that $tfpr^{cs}$ and $tfpr^{rr}$ are positively correlated and that their implied dispersion is comparable. These findings are important because if the assumptions in Hsieh and Klenow (2009) hold, they imply distortions are positively correlated with demand and technical efficiency shocks. One could argue for correlated distortions but this explanation raises the question why more distorted firms are more likely to hit by positive demand shocks and technical efficiency shocks. An alternative interpretation attributes dispersion to the effect of adjustment frictions. In the presence of such frictions, plants do not adjust production factors instantaneously and therefore high-productivity plants will exhibit high revenue productivity estimates. FGHW provide more details on this narrative.

It is worth noting that many other factors are potentially important sources of revenue productivity dispersion over and above physical productivity dispersion. FGHW show that the dispersion in revenue residuals depends on both dispersion in physical productivity and in demand shocks. In addition, the available empirical evidence in Foster, Haltiwanger, and Syverson (2008) suggests that physical productivity dispersion is higher than revenue productivity dispersion, which reflects the inverse relationship between productivity and prices. But Foster, Haltiwanger, and Syverson (2008) also highlight that demand shocks exhibit high dispersion relative to physical productivity dispersion. Hsieh and Klenow (2009) highlight the role of distortions in generating dispersion in revenue productivity. Others emphasize the role of frictions such as overhead factor costs (see, e.g., Bartelsman, Haltiwanger, and Scarpetta (2013)) and adjustment frictions (see, e.g., Asker, Collard-Wexler, and De Loecker (2014)).[18]

5.2 Growth and survival

In this section, we explore whether one of the most important predictions from standard models of firm dynamics is robust to the way productivity is estimated.[19] These models share a common ingredient: firms decide on exit or growth upon learning their ex ante uncertain productivity level. That is, firm dynamics are determined endogenously and firms' decisions are based on a firm-specific productivity shock. A common prediction of these models is that more productive firms are more likely to grow and survive than their less productive competitors. Empirical work on the connection between growth and productivity also relies on the results of these models. A study by FGH is a recent example of the research in this context. A comprehensive survey of the literature from the past decade can be found in Syverson (2011).

We now turn to our own empirical analysis of the relationship between productivity, growth and survival. Our discussion is centered on whether the most commonly found patterns hold across productivity estimators. We build upon the existing literature concerning the properties of productivity dynamics and test the robustness of these predictions using simple regression

[18] A related point concerns the sensitivity of dispersion measures to imputation. Recent research found that dispersion analyses based on imputed data may underestimate true productivity differences. We explore this question in more detail in appendix A.5.2. Our results confirm that accounting for imputation implies greater dispersion. Dispersion may be sensitive also to the homogeneity assumption one must maintain in order to be able use regression-based estimation methods. One can argue that plant-specific elasticities better capture technological differences between establishments and also affect productivity differences across plants. Our evidence, presented in detail in appendix A.5.3, shows that productivity dispersion is about two times higher if we allow for such heterogeneity. However, other results suggest some of this increase is noise.

[19] The most influential theories of firm dynamics are described in the classic models by Jovanovic (1982), Hopenhayn (1992), Ericson and Pakes (1995).

models linking a set of outcomes to productivity and plant-level controls. We follow FGH when considering the relationship between productivity and growth of all establishments, exiters and incumbents separately. This approach is justified by theoretical and empirical considerations. First, the basic models of firm dynamics themselves analyze these margins separately. Second, earlier empirical research found that there are differences in the productivity levels of continuers, entrants and exiters.[20]

Equation (4) describes our empirical specification, analogous to that in FGH:

$$Y_{i,t+1} = \gamma_1 \omega_{it} + \gamma_2 \theta_{state} + \gamma_3 \theta_{year} + \gamma_4 \theta_{size} + \gamma_5 u_{s,t+1} + \epsilon_{i,t+1}, \tag{4}$$

where $Y_{i,t+1}$ is the outcome of interest such as growth between t and $t+1$, ω and u denote a plant-level measure of productivity and state-level measure of change in unemployment from t to $t+1$. The θ-s denote state-, year- and establishment-size controls. i, t and s index plants, time periods and states, respectively. This specification differs from the one in FGH in that age effects, the Great Recession dummy and its interactions are omitted for simplicity.[21]

Table 5 shows $\widehat{\gamma}_1$ from equation (4) using our sample of the 50 most populous industries. Each row lists the effect of productivity on a specific outcome, shown in the row header. The three outcomes are: employment growth among all establishments, the probability of exit, and employment growth among continuers. The columns show results under productivity estimator variants. For example, the first entry in column 1 says that a plant is estimated to grow approximately 0.16% faster if it is 1% more productive under an OLS-based estimator. All other entries are analogous. Point estimates suggest there are non-trivial differences in the productivity effect. For example, the first entry in column 2 shows that the OP-based productivity effect is less than half of the OLS-based one. The difference between the results of these two estimators encompasses the variation in coefficients, ignoring for now WLPE and WLPM. Despite these non-negligible differences, the estimates support the earlier finding that more productive plants grow significantly faster than their less productive competitors. The estimates in row 3 show that low-productivity establishments are significantly more likely to exit than high-productivity establishments.

The results in the Panels B and C highlight the sensitivity of some methods to changes in sample definition. Panel B shows results when elasticities are estimated pooling data from all 4-digit industries within the same 3-digit industry. Pooling amounts to allowing stricter assumptions about the homogeneity of elasticities, which may be justified in order to increase precision. Comparing estimates between the top and middle panels suggests that some methods are sensitive to such changes. For instance, the absolute values of OP-based coefficients of all three regressions increase indicating that pooling may be beneficial because if there is no statistical association between two variables, their partial correlation coefficient would tend towards zero. Other estimators seem less prone to such changes and remain in a comparable range. WLPM yields the counter-intuitive result that more productive plants are less likely to grow and more likely to exit. Increasing sample size by pooling data does not reverse this result. However, if we drop industries with negative elasticities (see Panel C), the signs appear to be more in line with other results. In addition, the absolute values of point estimates also

[20] See, for example, Baily, Hulten, and Campbell (1992), Foster, Haltiwanger, and Krizan (2001), Foster, Haltiwanger, and Krizan (2006).

[21] We follow FGH by using the integrated LBD with the ASM data for this analysis. The ASM provides the distribution of plant-level productivity in any given year and the LBD provides the growth and survival outcomes for the full set of plants in the ASM in that year between t and t+1.

Table 5: The effect of productivity on outcomes, sample is 50 largest industries. Outcomes are: employment growth among all establishments (row 1), exit (row 2), employment growth among continuers (row 3).

	OLS	OP	LPGR	WLPE	WLPM	GA
Panel A. 4-digit elasticities						
overall growth	0.163***	0.072***	0.139***	0.06***	-0.006***	0.190***
exit	-0.050***	-0.018***	-0.046***	-0.020***	0.002**	-0.064***
conditional growth	0.068***	0.039***	0.053***	0.022***	-0.003***	0.067***
Panel B. 3-digit elasticities						
overall growth	0.190***	0.155***	0.152***	0.100***	-0.017***	0.183***
exit	-0.066***	-0.048***	-0.030***	-0.030***	0.005**	-0.062***
conditional growth	0.061***	0.065***	0.071***	0.043***	-0.008***	0.064***
Panel C. 3-digit elasticities, industries with negative or non-estimable elasticities dropped						
overall growth	0.197***	0.153***	0.164***	0.104***	0.076***	0.183***
exit	-0.069***	-0.044***	-0.030***	-0.036***	-0.028***	-0.062***
conditional growth	0.064***	0.072***	0.074***	0.035***	0.023**	0.064***

Estimates are taken from regressions of three outcomes (employment growth among all establishments, exit, and employment growth among continuers) on a plant-level measure of productivity (columns), a state-level measure of unemployment growth, year-, sizeclass- and state-fixed effects, see equation (4). Standard errors are clustered at the state level.
** and *** denote 5% and 1% significance levels, respectively. All regressions are based on trimmed productivity distributions. Sample size information can be found in table A17. Results for two additional industry sets can be found in tables A18-A19.

increase and/or standard errors decrease, despite the decrease in sample size.[22] These findings are intuitive. The OP-example highlights a case where there is better chance of estimating plausible elasticities by pooling data from neighboring industries. In such cases, sacrificing heterogeneity in revenue function coefficients by pooling data amounts to replacing less informative observations with more informative ones. WLPM is an example where only pooling and also dropping 3-digit implausibles help exclude less informative observations.[23]

To sum up, our estimates show positive (negative) and significant association between productivity and growth (exit), irrespective of how productivity is estimated. Some methods yield outlier elasticities that weaken the estimated relationship between productivity, growth and survival. In these cases, increasing sample size by using broader industry definitions yields results that line up better across methods.[24] The similarity of conclusions discussed in this section

[22]More information about sample size can be found in table A17.

[23]We carried out similar exercises using other samples from our default industry set. Without discussing them in detail, we note that a combination of using 3-digit elasticities and dropping industries with implausible elasticities yields expected growth and exit coefficients under all estimation methods in all the samples we considered. See tables A18-A19 for more details.

[24]A related question concerns the effects of allowing for not less but more heterogeneity in elasticities. Appendix A.5.3 shows that using productivity numbers that are based on plant-specific elasticities result in significantly higher productivity dispersion. The increase also implies weaker relationships between productivity, growth and survival. We interpret the latter finding as indirect evidence that the increased dispersion is due at least partly to measurement error implying that using plant-level shares is unlikely to be optimal. The effect of imputation on these coefficients is similar. However, despite attenuated coefficients, the main conclusions hold, see appendix A.5.2 for more details.

is a good example of the reasons why researchers have not focused on the differences between estimation methods. Finally, we return to the consequences of using revenue productivity as opposed to physical productivity in these exercises. Foster, Haltiwanger, and Syverson (2008) find that the marginal response of exit to revenue productivity is actually larger than that for physical productivity. They show that this is because revenue productivity includes both the effects of physical productivity and demand effects. We anticipate that similar remarks likely hold in this context.

5.3 Structural decompositions of aggregate productivity growth

In this section, we examine whether the estimation method affects the results from structural decompositions of aggregate productivity growth (APG). Productivity decompositions are identities that parse APG into components that are assumed to capture different sources of growth. Numerous papers explored how APG is tied to the behavior of microeconomic agents; some examples are Olley and Pakes (1996), Baily, Bartelsman, and Haltiwanger (2001), Petrin and Levinsohn (2012) and PWR. The concrete decompositions in these papers are different in many respects, but there is a common thread: they are all based on the general idea that APG sources either from the productivity gains within firms or the more efficient allocation of factor inputs between firms. We use a recent decomposition proposed by PWR primarily because the role of elasticities is made explicit in their model. They show that the contribution to APG by reallocation is a weighted average of input growth rates where the weights are determined by the difference between the marginal revenue product and marginal cost of inputs. The elasticities affect reallocation via marginal revenue products. Another nice property of PWR's approach is that the measure of APG itself does not depend on the way productivity is estimated because APG is defined as the growth in final demand in excess of capital and labor growth.[25]

Table 6 summarizes our results. Panel A shows the elements of the APG definition in PWR. The annual average growth rate for labor (-0.3%) in our sample is similar to that in PWR (-0.2%) but our value added and capital growth rates are smaller. These differences are due partly to measurement differences and partly to an issue with PWR's source data for implicit deflators, see the notes to table 6 for more details. However, value added growth is closer to PWR's measure if we ignore the first years of ASM panels, where calculating aggregate growth is problematic, see the rightmost columns in Panel A of table 6. Panel B lists the results of decomposing APG ignoring the first years of ASM panels. Each row corresponds to the contribution by a distinct component, calculated as an annual average between 1977 and 1996. The contribution of reallocation is shown in the row labeled as Total RE. Our estimate of the annual contribution by reallocation falls between 0.7-1.3 percentage points. These numbers imply that approximately 35-70% of annual APG is attributed to reallocation, depending on the chosen estimation method. The contribution of within-plant growth, shown in the row labeled as Within, is at most 1.4 percentage points, about 74% of total APG.[26] The variation in these results is substantial. For example, if we estimate elasticities using the WLPE or WLPM, we conclude that the engine of growth is reallocation. However, choosing LPGR would imply that the majority of APG is attributed to within-plant dynamics. These results are

[25]Since final demand is not observed, PWR measure its growth using value added. This approximation is exact only at the level of the total economy. It is therefore important to realize that by implementing the decomposition for a subset of plants we compute the contribution of the subset to APG, not the APG of the subset. To calculate the subset's exact APG, we would have to observe final demand for that subset.

[26]Including the first years of ASM panels yield similar results for reallocation and somewhat lower contributions by within-plant growth.

Table 6: Aggregate productivity growth and its decomposition. Annual averages calculated between 1977 and 1996 using ASM continuers.

Panel A. APG definition (%)

	All years		Ignoring first years*	
	Our sample	PWR	Our sample	PWR
VA	1.9	2.3	1.4	1.6
Capital	0.0	0.3	0.0	0.4
Labor	-0.3	-0.2	-0.5	-0.4
APG	2.2	2.2	1.9	1.6

Panel B. Annual average contributions by components, ignoring first years.

	OLS	OP	LPGR	WLPE**	WLPM**	GA	PWR
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Total RE	0.8	0.7	0.7	1.3	1.3	0.5	2
Capital RE	0.2	0.2	0.3	0.8	0.6	0.1	0.9
Labor RE	0.0	0.0	-0.1	0.0	0.0	0.1	0.4
Materials RE	0.4	0.4	0.4	0.5	0.5	0.3	0.4
Energy RE	0.2	0.1	0.1	0.0	0.1	0.0	0.3
Within	0.8	0.7	1.4	-0.1	-0.1	1.1	-0.1
Fixed costs	-0.3	-0.5	0.2	-0.8	-0.7	-0.3	0.2
σ_{RE}	0.8	0.5	0.7	0.9	0.8	1.2	1.7
σ_{Within}	2.3	2.8	3.1	2.5	2.5	2.4	2.7

*Measuring growth in first years of ASM panels is problematic because weighted growth rates are based on only large plants since growth rates do not exist for establishments just rotated in. **Instruments as in PWR: second and third lags. The last column in each panel is based on tables 1, 2 and 3a in PWR.
PWR generate data on non-production worker hours using variation in the average number of non-production workers and assuming a 40-hour working week and 50 weeks, while we estimate total hours as a function of production worker hours and the ratio of the total wage bill to the wages of production workers. The difference in capital growth is explained by a labeling issue in PWR's source data for implicit deflators. As a last point, we note there is a small difference in the way we calculate value added. While we use energy deflators to calculate constant-dollar energy costs, PWR deflate energy costs together with other intermediate inputs using material deflators, which may affect aggregate growth rates.

important because they highlight the fact that we may arrive at different conclusions about the relative importance of reallocation and within-plant growth depending on what estimator we use. It is important to emphasize that, despite the numerical differences mentioned above, all estimators imply positive average contributions for reallocation. We can interpret this as indication that reallocation was productivity-enhancing in U.S. Manufacturing industries between 1977-1996. This is in line with both existing theories of firm dynamics and our intuition about well functioning market economies.[27]

In the empirical implementation above, we followed PWR in assuming homogenous products and price taking behavior. Under these assumptions, revenue elasticities are equal to factor elasticities. However, this equivalence does not hold if prices are endogenous. In appendix A.4, we illustrate how marginal revenue products are affected by both factor elasticities and demand characteristics under endogenous prices and show that in this environment the correct measure of the difference between marginal revenue products and marginal costs is obtained using revenue elasticities in the decomposition. We assess the empirical relevance of the distinction between factor elasticities and revenue elasticities by evaluating decomposition results

[27]The timeseries underlying table 6 indicate that the contribution of reallocation is positive in the majority of years which reinforces what our annual average results suggest. We also note that the time series standard deviation of the total reallocation contribution is small relative to the within-plant productivity contribution.

with and without controlling for demand effects during estimation, both summarized in table 7. The main finding is that accounting for demand characteristics implies marginally lower contributions by reallocation (0.6 instead of 0.7) and much lower contributions by within-plant growth (0.1 instead of 0.7) see table 7 for more details. These patterns are consistent with our theoretical findings, see Appendix A.4 for details.

Table 7: The effect of controlling for demand characteristics in PWR's decomposition of APG.

	no demand	demand
Total RE	0.7	0.6
Capital RE	0.2	0.2
Labor RE	0.0	0.0
Materials RE	0.4	0.2
Energy RE	0.1	0.2
Within	0.7	0.1
Fixed costs	-0.5	-1.0

Results are based on OP and using data between 1977-1996, ignoring first years of ASM panels. Entries in the column labeled 'no demand' are identical to those in column 2 of Panel B in table 6. The column labeled 'demand' shows results under joint estimation of revenue elasticities and demand parameters, see section 4.

6 Concluding remarks

Researchers have been using a variety of methods for estimating productivity at the firm-level. Absent data on prices and quantities, these methods yield what have become known as revenue productivity measures. How these measures are related to physical productivity depends on the assumptions about the environment in which establishments operate. It is perhaps less recognized that the differences across estimation methods have important consequences for interpretation. One such difference concerns revenue function estimates: while cost-share-based coefficients are, in principle, equivalent to factor elasticities, regression-based estimates equal factor elasticities only under strict assumptions about product markets. This implies that revenue residuals are conceptually different under these two broad approaches.

Our objective is to assess the empirical importance of such distinctions in the context of basic stylized facts of the empirical literature. We find that the estimation methods affect revenue elasticity distributions, especially that of capital. Our structural demand analysis, admittedly restrictive, suggests this variation implies non-trivial differences in returns to scale across methods. One needs to take this variation into account when answering questions related to the magnitude of elasticities. The variation in elasticities maps into differences in within-industry productivity dispersion. Our estimates indicate that an establishment at the 75th percentile generates between 25%-50% percent more revenue with the same amount of inputs than a plant at the 25th percentile. This is an average calculated over industries and time, which suggests one must exercise caution when using industry-specific dispersion measures for economic analyses where small changes may make a big difference. In addition, the ranking of plants by productivity within industries is also sensitive to the estimation method. However, these results also confirm that there is enormous heterogeneity in establishments' productivity levels, which is in line with what recent microeconomic research found.

Canonical models of firm dynamics describe growth and survival decisions of plants' as functions of idiosyncratic productivity shocks. The main prediction from these models is that plants with positive shocks expand while plants with negative shocks downsize or exit. Our

data confirms these predictions: we find that more productive establishments grow significantly faster and are more likely to survive than their less productive competitors. However, outliers in factor elasticities that arise more frequently from some methods non-trivially impact the quantitative marginal effects of productivity on growth and survival.

Does the choice of estimation method have implications for the relative importance of the sources of aggregate productivity growth? We find that although some methods seem to be more sensitive than others and the relative importance of reallocation and within-plant growth depends on the estimator, the main conclusions of the decomposition literature hold under different estimator variants. Reallocation is productivity enhancing and within-plant growth seems to be more important for cyclical fluctuations. However, the quantitative details differ substantially. Some methods attribute the lion's share of aggregate productivity growth to reallocation while others imply that within-plant growth is more important. In addition, we find that controlling for demand effects has implications for the relative importance of reallocation and within-plant growth.

In sum, it is important to understand when the devil is in the details. One devil that may remain in the details is the impact of heterogeneous and endogenous plant-level product prices. The results from our - admittedly restrictive - demand analysis can be used to make inference about factor elasticities and returns to scale. However, without plant-level data on prices and/or quantities, the effects of prices on other key stylized facts are difficult to quantify. We have commented on the likely impact of endogenous demand side factors throughout but it would be of interest to consider this issue in more depth. We think that exploring the role of endogenous demand side factors in the current context will require comparing and contrasting approaches that include direct measures of prices and quantities (for the limited number of products with such information) with methods that impose strong functional form assumptions (i.e., isoelastic demand structures) to deal with these issues.

A more general conclusion from our analysis is that despite existing conceptual differences, available revenue productivity measures tell broadly consistent stories about the most important aspects of firm dynamics. This is a finding especially relevant in the context of the recent literature on misallocation. Hsieh and Klenow (2009) show that, under isoelastic demand and constant returns to scale technology, dispersion in cost-share-based productivity measures, or tfpr_i^{cs}, reflect the effect of distortions. On the other hand, FGHW show that regression based revenue residuals, or tfpr_i^{rr}, are functions of physical productivity and demand shocks. Under the assumptions in Hsieh and Klenow (2009), our results about tfpr_i^{cs} and tfpr_i^{rr} imply that distortions are positively correlated with demand and technical efficiency shocks. One could argue for correlated distortions but this explanation raises the question why more distorted firms are more likely to hit by positive demand shocks and technical efficiency shocks. An alternative interpretation attributes dispersion to the effect of adjustment frictions. In the presence of such frictions, plants do not adjust production factors instantaneously and therefore high-productivity plants will exhibit high revenue productivity estimates. The previous result demonstrates how the different assumptions affect interpretation and highlights the fact that users of these measures should carefully consider the assumptions underlying revenue function estimates when using and interpreting revenue productivity measures in subsequent analyses.

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A Appendix

A.1 Remarks on proxy methods

The details of the procedures described in Olley and Pakes (1996) and Levinsohn and Petrin (2003) have been discussed in the literature, see for example Ackerberg, Caves, and Frazer (2015). Caveats notwithstanding, these procedures have become widely used among practitioners in the past decade. The available versions of the empirical implementation differ according to the production function type and the numerical procedure used to minimize the objective function.

For example, OP's Stata commands are based on nonlinear least squares; while LP's implementation offers three options to minimize the GMM criterion function. If the dependent variable is value added then the golden section search algorithm is applied.[28] For output-based specifications, either a gradient-based or a grid search based routine may be chosen. Both the gradient-based and the golden section search guarantee to find optimum points if the objective function is unimodal. If the criterion function has multiple modes, grid search can be used to confirm global optimum but often at the cost of computational burden, especially if we are to estimate elasticities for many industries.[29] Our results in section 4 suggest these choices entail non-trivial differences in the distributions of elasticities.

Table A1: Variants of LP.

Method	Description	Dependent variable	Proxy	Estimator	Numerical procedure*
LPVA	Levinsohn and Petrin (2003)	Value Added	Materials	GMM	GSS
LPNL	Levinsohn and Petrin (2003)	Output	Materials	GMM	NL
LPGR	Levinsohn and Petrin (2003)	Output	Materials	GMM	GR
LPGSS	Levinsohn and Petrin (2003)	Output	Materials	GMM	GSS

*NL: gradient-based technique, GSS: Golden Section Search, GR: Grid Search.

A.2 Random assignment of industry codes

As described in Section 3.2, we correct for the 1972 to 1987 SIC change in our third analysis sample by following PWR by assigning the observed 1987 SIC code to the 1976-1986 observation for any establishment observed between 1987-1996. However, we deviate from their approach for cases only observed prior to 1987. If a plant is not assigned an industry code in the first step, we apply a random assignment procedure. The basic idea of the random assignment procedure is to choose from among plants such that the share of reassigned plants matches the appropriate share in the concordance. Randomness is necessary to ensure the procedure is not dominated by a few large establishments. As an illustration, suppose the mapping says 10% of industry i's (SIC 1972) total value of shipments should be mapped into industry j (SIC 1987). First, we compute the time-average of each plant's share in the shipments of industry i and then we randomly sort them by these averages. Next, we calculate the cumulative sum of shares and find the first n plants for which the sum does not exceed 10%. These establishments are classified in industry j.

Table A2 shows frequency counts from the assignment. Panel 1 summarizes the initial sample. About 66% of establishments show up in the years between 1987-1996, the remaining observations need industry assignment (about 34%). Panel 2 shows statistics about instances where we observed a switch in the industry identifier. Our assignment procedure implies that approximately 29% of the 130 thousand original switching instances disappear. As a cross-check, we compared the average shares our procedure implies to those in the crosswalk. The results of this latter exercise, not shown here, suggest that random assignment approximately replicates mappings in available concordances.

[28]The idea of GSS is successively narrowing the range of values inside which the extremum is known to exist. The name of the algorithm sources from the fact that the procedure maintains the function values for triples of points whose distances form a golden ratio.

[29]Estimating the 459 LPGR β-sets for the APG comparisons of section 5.3 took about 28 days given our computing resources.

Table A2: Descriptive statistics on industry assigment.

Panel 1. Distribution of plants, by year.

Observed year	Frequency (1000)	Percent
1987	1156	51.3
1988	5.1	0.2
1989	7.6	0.3
1990	9.8	0.4
1991	3.5	0.2
1992	230	10.2
1993	20.8	0.9
1994	30.3	1.3
1995	19.4	0.9
1996	15.1	0.7
1987-1996	1499	66.5
1972-1986 (random assignment)	754	33.5
Total	2252	100

Panel 2. The effect of industry assignment on switchers.

Original industry	Assigned industry: no switch	switch	Total
frequency (1000)	38	92	130
percent	29	71	100

Panel 1 shows frequency counts of time periods which were used to assign plants into industries. Panel 2 breaks down switching instances (plants with 2 or more SIC codes in their time series) under the original classification. The first entry in the last row says that 29% of the switching instances in the original classification system disappear under random assignment.

A.3 Empirical properties of estimators

The distributions discussed in section 4 are based on elasticities from the 50 most populous 4-digit industries. While 50 observations may seem sufficient to estimate cross-industry elasticity distributions, it also means estimates are based on varying sample size. This is important because the estimates from smaller industries are more likely to be less precise. Table A3 shows that the sample size drops by more than 50% in the 10th, and by 80% in the 50th most populous industry. What is the consequence of such variation? Are some methods more likely to yield non-positive $\hat{\beta}$-s than others?

In order to answer these questions, we re-estimate the above elasticity distributions using all 459 industries and count the cases with positive, zero, and negative elasticities for every method.[30] All methods result in positive $\hat{\beta}_l$ and $\hat{\beta}_m$ in most industries (table A4, columns 6 and 9). Negative β_k and β_e estimates are generally more likely to occur, as shown by smaller percentages in columns 3 and 12. There are differences across estimation methods, as well. For example, LPGR always delivers positive $\hat{\beta}_k$, while other methods yield negative $\hat{\beta}_k$ with a positive probability. OP's algorithm stops in 18% of the industries (columns 2, 5, 8 and 11) due to the lack of information on exiters and LPNL yields zero $\hat{\beta}_k$ in 16% of the industries (column 2). We obtain negative $\hat{\beta}_e$-s with especially high probability when using WLPE (column 10).

Can the variation in sample size explain these patterns? Comparing the average number of plant-year observations in the problematic group (negative, zero or non-estimable) to that of the positive group suggest the answer is at least partially yes. Problematic industries are generally

[30]The sample for this exercise is described in more detail at the end of section 3.2.

Table A3: Descriptive statistics of 50 largest industries ordered by the within-industry number of plant-year observations between 1976-1996. The industry classification contains SIC 1987 industry codes concorded 1-to-1 SIC 1972 and NAICS 1997.

(1) rank	(2) SIC 1987	(3) N(1000)	(4) $\overline{T}_i$	(5) N_j
1	2711	39	9.9	3.9
2	3273	37	10.1	3.7
3	2411	37	4.8	7.6
4	3441	32	11.8	2.6
5	2653	30	22.1	1.3
10	3442	16	12.4	1.3
20	3231	12	11.9	1
50	3613	8	18	0.4
1-50 total		710	11	67

Column 1: rank; column 2: SIC 1987 code; column 3: number of plant-year observations in thousands; column 4: average number of observations per plant; column 5: number of plants in thousands.

smaller, their average size is between 26-70% of the positive group (see table A5 for more details). For example, in the industries where OP stops the average number of observations is less than half of that in the positive group. However, we find two distinct cases where erratic estimates are unlikely to be related to sample size. First, the zero-$\hat{\beta}_k$ group for LPNL is of very similar size as the positive group (second entry in column 2 of table A5). It is likely that LPNL's gradient-based numerical procedure stops at a local optimum point at zero. The second exception is the negative $\hat{\beta}_e$ group for WLPE where average sample size is similar to that in the positive group (column 7 of table A5). These results suggest outliers may arise more frequently from some methods than others. In practice, what should be done with negative elasticity estimates that emerge from some methods? One approach would be to exclude industries with negative elasticity estimates since such estimates are implausible but that would raise issues of selection bias. An alternative is to make stronger assumptions of homogeneity – for example, assuming plants within 3-digit SIC industries share the same factor elasticities. We explore this approach in section 5.2 and show that this approach leads to more plausible elasticity estimates and implied firm dynamics in many cases.

Table A4: The number of industries with positive, negative and non-estimable elasticities, in percent of the total number of industries. Results are based on 459 4-digit industries between 1972-2010, see section 3 for more details.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	$\widehat{\beta}_k$			$\widehat{\beta}_l$			$\widehat{\beta}_m$			$\widehat{\beta}_e$		
	$-$	0* or n.e.**	$+$	$-$	0 or n.e.	$+$	$-$	0 or n.e.	$+$	$-$	0 or n.e.	$+$
OLS	2		98	2		98			100	4		96
OP	3	18*	79	2	18*	80		18*	82	3	18*	79
LPVA	4		96	3		97						
LPNL		16**	84	4		96			100	6		94
LPGSS	8		92	4		96			100	6		94
LPGR			100	4		96			100	6		94
WLPE	10		90	6		94			100	32		68
WLPM	16		84	7		93	2		98	11		89

0*: $\widehat{\beta}$ is zero. *n.e.***: $\widehat{\beta}$ could not be estimated.
For example, the first entry in column (2) says OP delivers error in 18% of industries. This happens because the algorithm stops in industries with insufficient information on exit (or investment). The second entry in column (2) says gradient-based optimization in LPNL yields $\widehat{\beta}_k = 0$ in 16% of the industries.

Table A5: Average number of within-industry observations by groups, as in table A4.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	$\widehat{\beta}_k$		$\widehat{\beta}_L$		$\widehat{\beta}_M$		$\widehat{\beta}_E$	
	-	0 or n.e.	-	0 or n.e.	-	0 or n.e.	-	0 or n.e.
OLS	0.59		0.46				0.32	
OP	0.56	0.39	0.58	0.39		0.4	0.42	0.39
LPVA	0.26		0.29					
LPNL		0.96	0.55				0.42	
LPGSS	0.35		0.55				0.42	
LPGR			0.55				0.42	
WLPE	0.4		0.56			0.14	0.93	
WLPM	0.7		0.52		0.61		0.44	

All entries are calculated respectively as $\bar{N}_-/\bar{N}_+$ and $\bar{N}_{0 \text{ or n.e.}}/\bar{N}_+$, where $\bar{N}_-$, $\bar{N}_{0 \text{ or n.e.}}$ and $\bar{N}_+$ denote the average number of observations in industries with negative, zero or non-estimable and positive $\widehat{\beta}_j$. See also the notes below table A4.

A.4 PWR's decomposition under downward sloping isoelastic demand

In contrast to section 5.3, assume that firms are not price takers in output markets but they operate under isoelastic, downward sloping demand conditions as in Foster, Grim, Haltiwanger, and Wolf (2016b), so the inverse demand function can be written as $P_i{=}P(Q/Q_i)^{1-\rho}\xi_i$, where ξ_i is an idiosyncratic demand shifter. We maintain the assumption that firms are price takers on the market for intermediate inputs M_j. Under these assumptions, changes in the allocation of inputs will have an effect on the marginal revenue products of inputs. The total differential of the revenue function is given by

$$\begin{aligned} d(P_iQ_i) &= PQ^{1-\rho}\xi_i\rho Q_i^{\rho-1}dQ_i \\ &= PQ^{1-\rho}\xi_i\rho Q_i^{\rho-1}\left(\sum_k \frac{\partial Q_i}{\partial X_k}dX_{ik} + \sum_j \frac{\partial Q_i}{\partial M_{ij}}dM_{ij} + \frac{\partial Q_i}{\partial \Omega_i}d\Omega_i - dF_i\right). \end{aligned}$$

Summing the previous equation over firms gives the contribution by marginal revenue products, and subtracting the change in aggregate input expenditures (both primary and intermediate) and fixed costs yields APG:

$$\begin{aligned} &\sum_i\sum_k \left(PQ^{1-\rho}\xi_i\rho Q_i^{\rho-1}\frac{\partial Q_i}{\partial X_{ik}} - W_{ik}\right)dX_{ik} + \sum_i\sum_k \left(PQ^{1-\rho}\xi_i\rho Q_i^{\rho-1}\frac{\partial Q_i}{\partial M_{ij}} - P_j\right)dM_{ij} \\ &+ \sum_i PQ^{1-\rho}\xi_i\rho Q_i^{\rho-1}\frac{\partial Q_i}{\partial \Omega_i}d\Omega_i + \sum_i PQ^{1-\rho}\xi_i\rho Q_i^{\rho-1}dF_i. \end{aligned}$$

Under Cobb-Douglas technology, the k-th marginal revenue product can be written as $\frac{\partial P_iQ_i}{\partial X_{ik}}$ $=PQ^{1-\rho}\rho Q_i^{\rho-1}\xi_i\left[\frac{\partial Q_i}{\partial X_{ik}}\right]=\rho PQ^{1-\rho}\xi_iQ_i^{\rho}\frac{\alpha_k}{X_{ik}}$, and the reallocation term can be rewritten as

$$\sum_i\sum_k \left(\rho\left[PQ^{1-\rho}\xi_iQ_i^{\rho}\right]\frac{\alpha_k}{X_{ik}} - W_{ik}\right)dX_{ik}.$$

Noting that the term in the squared brackets gives the firm's revenues, the implication is that when translating contributions into growth rates, the same logic applies as in PWR. We obtain

$$\sum_i D_i \sum_k \left(\rho\alpha_k - s_{ik}^{\text{rev}}\right) d\ln X_{ik}, \tag{5}$$

which shows that under these assumptions the correct way to measure the difference between marginal revenue products and marginal costs is using revenue elasticities in the decomposition. The contribution of within-firm growth under endogenous prices defined by iso-elastic demand is obtained analogously. The final expression is given by $\sum_i \rho D_i d\ln\Omega_i$.

These results show that the main difference relative to PWR's decomposition is captured by the demand parameter ρ. They highlight the fact that under downward sloping iso-elastic demand ($0 < \rho < 1$), firms will take price effects into account when adjusting production inputs. Specifically, the effect of input reallocation is less than or equal to the effect under exogenous prices because revenue elasticities are smaller than factor elasticities. The contribution of within-plant growth is different because the revenue residual is different (joint estimation) and

also because the contribution of each industry is scaled by ρ.

What is the empirical relevance of accounting for demand during estimation? We answer this question using the tools introduced in section 4. In that section we showed that if we estimate revenue elasticities jointly with the demand parameter then there is an exact relationship between revenue elasticities and factor elasticities: $\rho\alpha_k = \beta_k$. In other words, using revenue elasticities is theoretically justified under these assumptions. However, revenue elasticities may be estimated also without accounting for demand effects, so the question is whether these matter for the basic results of the decomposition. As table 7 in section 5.3 shows, such distinctions can be relevant in empirical terms.

A.5 Robustness

We next examine the robustness of our results about productivity dispersion and growth and survival to concerns about imputation methods used in the underlying microdata and our assumption that elasticities are homogeneous within industries. In contrast to previous exercises, our approach here is to compare the same productivity measure across completed and non-imputed samples rather than different productivity measures in the same sample.

A.5.1 Imputation and plant characteristics

U.S. Manufacturing data collected by the Census Bureau is subject to item nonresponse where respondents answer some questions but not others. Such missing values are imputed by the Census Bureau. Recent research found that certain imputation methods can impact analyses that use completed data because imputation is non-random (see White, Reiter, and Petrin (2012), WRP hereafter). Some of the imputation methods employed use industry level data or fitted values from regression models implying that the variation in completed data can be assumed to be smaller relative to what observed responses would imply. One consequence of this is that within industry dispersion statistics based on completed data may be biased downward (see WRP for an example). The presence or direction of possible biases is less clear for the relationship between productivity, growth and survival for reasons we will outline shortly.

The effect of imputation on dispersion is easiest to illustrate by considering the results of regression-based imputation, which also happens to be one of the most frequently used methods at the Census Bureau to impute the components of productivity, see table A6 for more details.[31] Regression-based imputation amounts to substituting fitted values of a regression for values of the underlying distribution. That is, using the regression line, $E(y_i|x_i)$, instead of draws from the conditional distribution of y_i. This essentially means a collapse in variation in the data.

How can we approximate the unobserved conditional distribution of y_i? One way is to use a known distribution to simulate data from it. Another is to draw from the set of non-imputed observations with similar characteristics to the ones in the imputed sample, which is what classification and regression tree (CART) methods do (see WRP). Yet another approach is to use non-imputed observations only, which works if imputation is random. However, if the probability of imputation is correlated with plant characteristics, excluding imputed observations generates selection issues. Both earlier evidence and our analysis - as we will see shortly - indicate that imputation is not random in our data. Consequently, empirical models based on non-imputed data must take this issue into account otherwise selection renders results biased.

We address imputation in two steps. First, we use a logistic regression to describe the relationship between plant characteristics and the probability that productivity is calculated

[31] We do not discuss individual imputation models further but more details are available upon request.

Table A6: Estimated rates of imputation models within the most frequent impute types, as a percentage of the total number of observations, ASM-CM 2007. The most frequent impute types are based on linear regressions (flagged 'B') and/or historical (flagged 'H') information.

	tvs	ph	cm
ASM CM			
Overall imputation rate (fact)	28.5	58.8	41.1
Types: B or H flags (fact)	20.6	55.1	33.3
Models (estimate)*			
multivariate regression	4.1	12.4	6.3
univariate regression	15	44.8	26.6
historical	0.9	5.8	0.5
total	20	62.9	33.4
ASM cases only			
Overall imputation rate (fact)	24.1	39	34.3
Types: B or H flags (fact)	18.2	36.1	25.7
Models (estimate)*			
multivariate regression	10.5	14.4	14.4
univariate regression	8.1	13.5	11.6
historical	1	9.7	1.4
total	19.6	37.6	27.4

*Census data indicates the impute type but not the model and there are more than one impute models within a type. For example, within type-B imputes, both univariate and multivariate regression models are used. We estimate impute model instances by evaluating the restriction each impute model implies. These results show regression-based imputes are most common. In the ASM-CM, imputes are typically based on univariate regressions, while ASM imputes are much more likely to be based multivariate regressions.

using non-imputed data. Next, we use inverse propensity scores to weight observations in the non-imputed sample to calculate dispersion statistics and growth coefficients. To be specific, we estimate the following equation separately for each year between 2002 and 2010:

$$log\frac{p(X_{it})}{1-p(X_{it})} = X_{it}\theta_t + \epsilon_{it},$$

which amounts modeling the probability $Pr(I_{it} = 1|X_t) = E[I_{it}|X_{it}] = F(X_{it}\theta_t) + \epsilon_{it}$, where $F(x) = \frac{1}{1+e^{-x}}$ and I_{it} denotes an indicator variable equal to 1 if any of the components of plant-level productivity is non-imputed. The main components are: plants' total value of shipments (TVS), production hours (PH), salaries and wages (SW), production workers wages (WW), cost of parts (CP). X_{it} and θ_t denote a vector of controls and coefficients. Control variables X_{it} are included to capture plant characteristics: industry effects, employment size class, payroll deciles and age class fixed effects. We have 86 4-digit NAICS industries. To control for size, we defined 10 size classes based on employment (1-9, 10-109, 20-29, 30-49, 50-99, 100-149 150-249, 250-499, 500-999, 1000+) in addition to the payroll deciles. Finally, we classify establishments into 9 age classes (births, 1, 2, 3, 4, 5, 6-10, 11-15, 16+ years). As mentioned in the main text, data for these variables source from the LBD. We use the Census Bureau's impute flags to determine whether an item is imputed. More details on imputation procedures at the Bureau are available from the authors. See also table A1 in WPR.

Imputation rates differ across variables. The upper panel of table A7 shows that from among the main components of productivity, PH and CP are imputed the most, and SW the

least frequently. Imputation rates vary not only with variables but also sample definition.[32] The last row in the upper panel indicates imputation tends to be less frequent among ASM establishments. Overall, less frequent imputation implies almost 10-percentage-point smaller imputation rate for productivity (last column in the lower panel). This suggests it may be worth exploring the effects of restricting ourselves to ASM cases when estimating propensity scores. Therefore, we present results also for a scenario where only ASM establishments are included in the analysis.

Table A7: Imputation rates of the main components of productivity in the ASM/CM, as a per cent of plant-year observations between 2002-2010.

Average imputation rates (2002-2010)

	productivity	TVS	PH	SW	WW	CP	CM
ASM/CM*	59.1	28.1	42.6	14.2	29.2	42.6	37.5
ASM/CM/LBD**	58.3	27.6	42	14.1	28.5	41.7	36.3
ASM/LBD***	48.9	25.5	35.3	13.2	25.1	34	31.9

Yearly imputation rates of productivity, per cent of plant-year observations

	2002	2003	2004	2005	2006	2007	2008	2009	2010	Total
ASM/CM	68.8	45.8	42.6	43.8	49.9	69.2	53.2	52.4	53.9	59.1
ASM/CM/LBD	68	45.1	41.5	42.8	49.1	68.1	52.7	52	53.7	58.3
ASM/LBD	48.1					52.8				48.9

Productivity is considered imputed if at least one of its components is imputed by the Census Bureau. Components: total value of shipments (TVS - changes in inventories are not considered here); total hours, calculated as a product of production worker hours (PH) and the ratio of salaries and wages (SW) and production worker wages (WW); cost of materials (CM), calculated as a sum of the cost of parts (CP), resales (CR) and contract work (CW), but only CM and CP are included in the table. We excluded capital from the analysis because plants' time series on capital are created using the perpetual inventory method. *ASM/CM: ASM and CM combined. **ASM/CM/LBD: observations in ASM and CM for which we observe employment in the LBD. ***ASM/LBD: observations in ASM for which observe employment in the LBD.

Comparing size-, age-, and payroll distributions across non-imputed and completed samples suggests imputation instances are correlated with plant characteristics.[33] Productivity components are more likely to be imputed for smaller and younger establishments with less payroll. These three characteristics give a multitude of possible regressor sets for the probability model. We experimented with six of those ([1] employment size; [2] payroll; [3] employment size, payroll; [4] employment size, age; [5] payroll, age; [6] employment size, payroll, age) and found that the basic implications do not change. However, we also found that including all three variables provides a somewhat better fit[34] than any of the remaining five. Therefore, we present results based on propensity scores from a model with establishment size, payroll and age.

Figure A4 plots the point estimates from this model. We conclude that productivity data for smaller and younger plants with less payroll are significantly more likely to be imputed.[35] In

[32] We can measure imputation rates in the entire ASM/CM (row 1), restricting ourselves to using observations for which there exists size and age information in the LBD (row 2), in the ASM only (row 3).

[33] These results are not shown here.

[34] As measured by the AIC, not shown here.

[35] Point estimates are precise enough such that we can confirm the positive relationship between plant-size, -age and the probability of productivity being non-imputed. More details on point estimates and standard errors are available upon request.

section A.5.2, when calculating weighted dispersion measures and regression coefficients, we use the inverse of probabilities implied by these models as weights. As a final point, we mention that diagnostics indicate logistic regressions fit ASM establishments better than ASM/CM establishments.[36] The difference in the AIC is about a factor of 3 and 2 in 2002 and 2007, respectively.

A.5.2 Imputation, dispersion and growth

One may address imputation in a variety of ways. One approach is to drop imputed observations, but this results in selection bias if the probability of imputation instances is correlated with establishment characteristics. We take this approach below and attempt to correct for selection bias by using inverse propensity score weights discussed in section A.5.1. Another option is to use multiple imputation methods (e.g., the classification and regression tree method (CART) used by WRP) to improve on the methods that have been used to impute the plant-level data. Calculating our results in this manner is beyond the scope of this paper but we do compare the patterns of our findings to those in WRP.

We carry out two exercises to assess the effect of imputation. First, we compare dispersion in the completed sample to statistics which are based on non-imputed observations only. Second, we assess whether accounting for imputation affects our results on the relationship between productivity, growth and survival. We address imputation by first constructing a set of propensity score weights, which are inversely proportional to the probability of imputation and then using these weights to calculate weighted dispersion measures. Note that we use another set of weights to control for the fact that selection into the ASM is also non-random. We distinguish between the two sets by labeling non-impute weights as $ipw2$ and the ASM-weights as $ipw1$. Appendix A.5.1 describes how $ipw2$ was constructed. More details about $ipw1$ can be found in Foster, Grim, and Haltiwanger (2016a).

Figure A1 summarizes the results of the first exercise. We find that the interquartile range seems to be smaller in non-inputed data. This can be seen in figure 1(a) by comparing the dotted and solid lines. However, if we weight non-imputed observations by the composite weight [$ipw1$*$ipw2$], measured dispersion is higher.[37] This is an important finding because results using the CART multiple imputation method in WRP suggest dispersion measures based on imputed data tend to be smaller than those that take imputation into account.[38] If imputation causes a downward bias in dispersion, our weighting scheme corrects for it in the right direction. We note that imputation has similar effects on the standard deviation but the this measure seems to be less sensitive to these issues (see figure 1(b)).

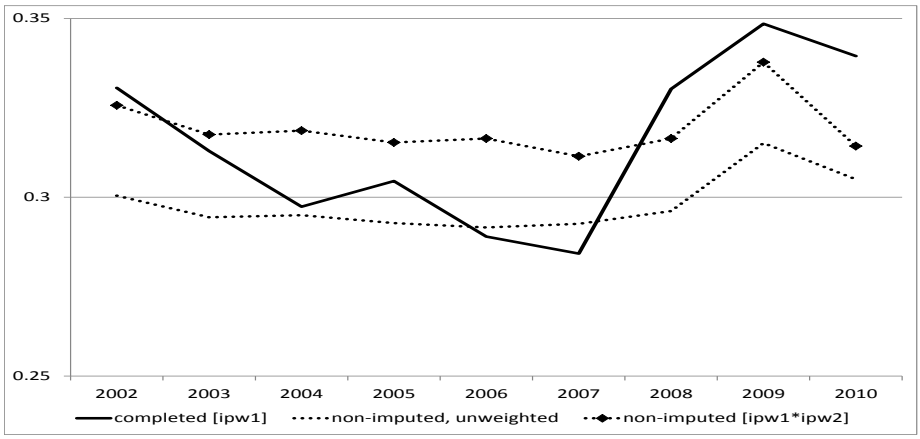
In the second exercise, we revisit the relationship between productivity, growth and survival and re-estimate the growth and exit regressions using both completed and non-imputed data between 2002 and 2010. The empirical model is the same as what we described in section 5.2. We consider three samples. In the first one, we use all observations from the CM.[39] In the second case, only non-imputed observations are included but our non-impute weighting scheme is not applied. In the last regressions we weight non-imputed observations by [$ipw1$*$ipw2$]. Table A8 summarizes our results. The effect of productivity on growth among all establishments (column 1) and on the probability of exit (column 2) seems smaller in the non-imputed sample,

[36] Available upon request.

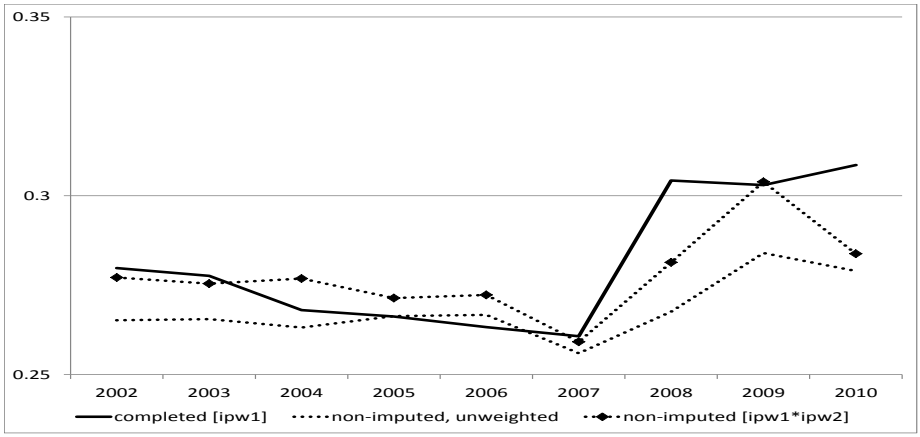
[37] Similar patterns hold for the 90-10 ratio, not shown here.

[38] See columns 1 and 2 of table 4 in their paper.

[39] Note that as the estimates in section 5.2 were calculated using $ipw1$ weights, all regressions in this exercise - based on either completed or non-imputed observations - are also weighted by $ipw1$.



(a) Inter-quartile range



(b) Standard deviation

Figure A1: Time series of average within-industry dispersion measures in various samples, ASM 2002-2010, all industries.

Solid and dotted lines denote statistics which are calculated using completed and non-imputed data, respectively. In the completed sample, weighted statistics are based on weights which account for the fact that selection into the ASM and CM is non-random ($ipw1$). In the non-imputed sample, the weights are a composite of $ipw1$ and a weight that is inversely proportional to the probability that the plant's productivity is calculated using non-imputed data ($ipw2$).

regardless of weighting. Point estimates of the effect on growth (exit) in the non-imputed sample are about half (third) the size of those in the completed data. We do not detect such attenuation among continuers (column 3). Such variation in the coefficients is by no means negligible, particularly for economic analyses where differences in magnitudes may matter a lot. But it is also familiar from earlier tables. Similarly to the results in section 5.2 our last exercise offers evidence that, at least in terms of sign and order of magnitude, this relationship is robust to imputation issues.

Table A8: The effect of productivity on outcomes between 2002-2010 in the ASM/CM, various samples within the ASM/CM, growth accounting based productivity.

sample	overall growth (1)	exit (2)	conditional growth (3)
		ASM-CM	
Completed [ipw1]†	0.165***	-0.065***	0.04***
Non-imputed [ipw1]††	0.083***	-0.019***	0.047***
Non-imputed [ipw1*ipw2]†††	0.079***	-0.021***	0.041***
		ASM cases only	
Completed [ipw1]	0.149***	-0.06***	0.035***
Non-imputed [ipw1]	0.086***	-0.022***	0.044***
Non-imputed [ipw1*ipw2]	0.086***	-0.024***	0.041***
Sample size (in thousands)			
		ASM-CM	
Completed data [ipw1]	594	594	570
Non-imputed data [ipw1]	263	263	258
Non-imputed data [ipw1*ipw2]	263	263	258
		ASM cases only	
Completed data [ipw1]	400	400	386
Non-imputed data [ipw1]	218	218	214
Non-imputed data [ipw1*ipw2]	218	218	213

Triple-asterisks denote 1% significance levels. †Weighted Census data [ipw1]: All observations, propensity score weighted, where the weight is inversely proportional to the probability that the plant is selected into the ASM/CM (see FGH). ††Non-imputed [ipw1]: non-imputed subset of the same sample. †††Non-imputed [ipw1*ipw2]: non-imputed subset of the sample, where observations are weighted by a composite propensity score, where $ipw2$ is inversely proportional to the probability that a plant's productivity is calculated using non-imputed data. Probabilities were estimated separately for each year and are based on industry-, size-, age-class and payroll-decile fixed effects.

A.5.3 Homogeneity, dispersion and growth

Throughout this paper, we assume that elasticities are homogeneous within industries and constant over time. Whether or not such an assumption is restrictive depends on at least two properties: the underlying within-industry differences in technology and sample size.[40] If there are within-industry differences in technology then pooling the data from an entire industry of establishments may be too restrictive. In this case, allowing for plant-level heterogeneity in the elasticities, empirical feasibility aside, better accounts for within-industry differences in factor intensities. On the other hand, pooling data may be necessary to increase sample size in order to

[40]It is not straightforward to test for the true degree of heterogeneity because the results on which we base inference about heterogeneity are endogenous to both the estimation method and the homogeneity assumption.

reduce finite-sample bias and increase precision.[41] A more general but equally important point in this regard is that pooling also implies results are less likely to be sensitive to measurement error, which is typically present in micro data.

In this section, we relax the homogeneity assumption and assess the consequences for productivity dispersion and growth and exit regressions. Figure A2 shows that dispersion is substantially higher if we allow cost-shares to vary across establishments compared to when it is constant within an industry. In particular, plant-specific shares increase the interquartile range by a factor of almost two, see the difference between the thin and thick lines in the figure. Results also indicate that allowing for time varying elasticities affects only the volatility of dispersion but not its level. In figure A2 this is shown by observing the difference between the thin dashed and solid lines. There is no significant difference in variation with industry-level shares implying the overall conclusion that the effect of time series smoothing is dwarfed by that of cross-section smoothing.

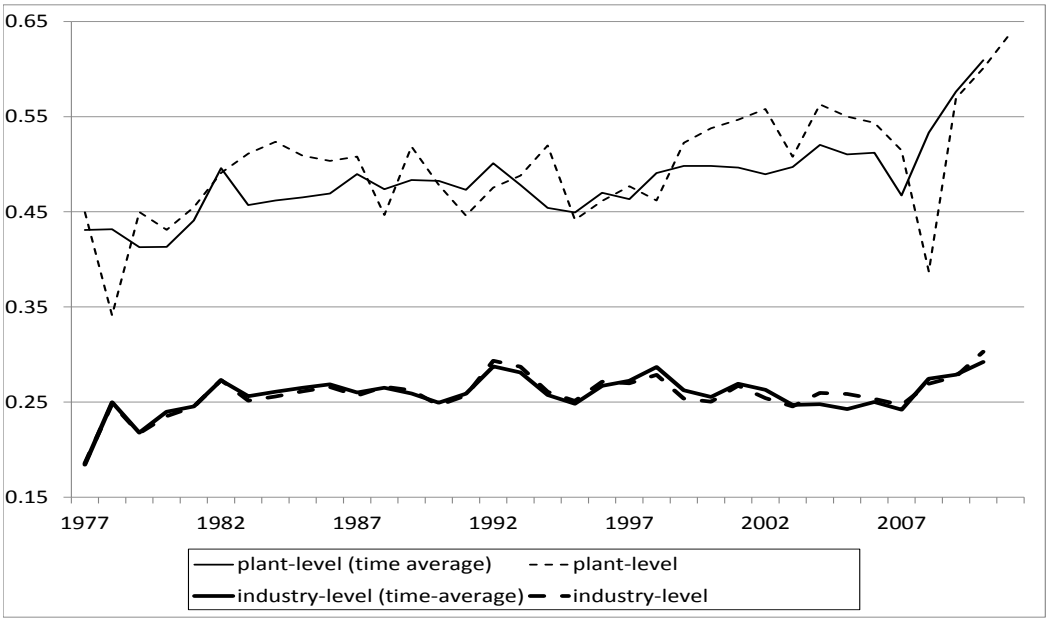


Figure A2: Within-industry interquartile range measures of growth accounting based log-productivity. Sample: 10 largest industries. Thin and thick lines denote dispersion measures based on shares calculated at the establishment- and industry-levels, using the growth accounting framework. Solid lines denote dispersion measures based on cost shares that are constant over time. Dashed lines denote statistics calculated using time-varying cost shares.

Both theory and earlier research[42] indicate that part of this increase in dispersion may be spurious. Our own analysis also offers indirect evidence that the increased variation is at least partly noise. Table A9 shows results from growth and exit regressions based on growth accounting productivity estimates. Comparing columns 1-2 shows that using time-varying industry-level shares leaves the effect of productivity virtually unchanged. In contrast, using plant-level shares (columns 3-5) reduces the magnitude of estimates, although they remain

[41] Moreover, for some of the estimators reviewed in this paper whether one pools data from different industries is not a matter of bias and precision but feasibility. See sections 4 and 5.2 for examples.

[42] See, for example, Syverson (2004).

statistically significant. The consequence of using plant-level shares is that the growth effect of productivity drops by almost 50% (columns 1 and 3) among all establishments (first row) or continuers (last row). Among continuers, an additional 25% of this effect disappears if we also allow for time series variation in the shares (columns 3-4). The effect on the exit probability is smaller, about 33%. We interpret this attenuation as a sign that the data contains more noise with respect to the relationship between productivity and growth. Note that constant plant-level shares imply a higher point estimate among continuers (column 3 relative to 4 and 5) suggesting there may be noise not only in the cross-section but also in the time series variation of establishment-specific cost shares. Our overall conclusion is that calculating cost shares at the plant level is unlikely to be optimal. Although plant-level shares may better capture within-industry differences in technology, they are also prone to measurement error. This is reflected in the smaller partial correlations between productivity, growth and the survival probability of establishments.

Table A9: The effect of growth accounting productivity variants on outcomes, 50 largest industries.

	Industry-level shares		Plant-level shares		
	(1)	(2)	(3)	(4)	(5)
	constant	time-varying	constant	time-varying	time-varying
	$[\bar{s}^j]$	$[\frac{s_t^j+s_{t-1}^j}{2}]$	$[\bar{s}_i^j]$	$[s_{it}^j]$	$[\frac{s_{it}^j+s_{it-1}^j}{2}]$
overall growth	0.190***	0.193***	0.114***	0.1***	0.096***
exit	-0.064***	-0.066***	-0.042***	-0.041***	-0.039***
conditional growth	0.067***	0.067***	0.032***	0.018***	0.018***

Outcomes are: employment growth among all establishments (rows 1-2), exit (rows 3-4), employment growth among continuers (rows 5-6). Estimates are based on the same 50 largest industries used to generate the results in table 5. The entries in column 1 above are identical to the last column of the first panel in table 5.

A.6 Additional Tables

Table A10: Conditional probabilities of an industry moving from quintile i (row) to quintile j (column) of the $\hat{\beta}_K/\hat{\beta}_L$-distribution, 50 largest industries.

	OP	1	2	3	4	5
LPGR	1	0.5	0.2	0.1	0.1	0.1
	2	0.3	0.3	0.2	0.2	0
	3	0.1	0.3	0.3	0.1	0.2
	4	0.1	0.1	0.3	0.2	0.3
	5	0.1	0.1	0.4	0.3	0
	GA	1	2	3	4	5
LPGR	1	0.1	0.2	0.3	0.2	0.2
	2	0.3	0.2	0.3	0.2	0
	3	0.3	0.2	0.2	0.2	0.1
	4	0.2	0.3	0.1	0.3	0.1
	5	0.1	0.1	0.1	0.3	0.4
	WLPM	1	2	3	4	5
LPGR	1	0.4	0.3	0.1	0.1	0
	2	0.2	0.1	0.3	0.2	0.1
	3	0.3	0.2	0.3	0.1	0.1
	4	0.1	0.3	0.2	0.2	0.2
	5	0.1	0.3	0.6	0	0

See notes to table 1 for legends. The first entry in the table says that half of the industries in the lowest quintile under LPGR (row) are also classified in the lowest quintile under OP (column). All other entries are analogous.

Table A11: Descriptive statistics of parameter estimates under isoelastic demand.

Panel A. Demand parameters.								
	$\hat{\beta}_q$		$\hat{\rho}=1-\hat{\beta}_q$		$\varepsilon_{P_i}^{Q_i}=\frac{1}{\rho-1}=-\frac{1}{\hat{\beta}_q}$		$\frac{1}{\rho}=\frac{1}{1-\hat{\beta}_q}$	
	OLS	OP	OLS	OP	OLS	OP	OLS	OP
mean	0.09	0.05	0.91	0.95	-11.83	-19.53	1.2	1.11
stdev	0.19	0.16	0.19	0.16	46.26	126.2	0.51	0.45
Panel B. Revenue elasticities: $\hat{\beta}_j$								
	$\hat{\beta}_K$		$\hat{\beta}_L$		$\hat{\beta}_M$		$\hat{\beta}_E$	
	OLS	OP	OLS	OP	OLS	OP	OLS	OP
mean	0.1	0.11	0.29	0.27	0.5	0.49	0.11	0.1
stdev	0.06	0.09	0.1	0.09	0.13	0.13	0.05	0.05
Panel C. Factor elasticities: $\hat{\alpha}_j=\hat{\beta}_j/\hat{\rho}=\hat{\beta}_j/(1-\hat{\beta}_q)$								
	$\hat{\alpha}_K$		$\hat{\alpha}_L$		$\hat{\alpha}_M$		$\hat{\alpha}_E$	
	OLS	OP	OLS	OP	OLS	OP	OLS	OP
mean	0.13	0.13	0.35	0.31	0.58	0.54	0.14	0.11
stdev	0.12	0.14	0.2	0.2	0.23	0.23	0.12	0.07

We assume $P_i{=}P(Q/Q_i)^{1-\rho}$ in the estimation. The parameters related to demand exhibit large standard deviation which is a sign that these parameters are poorly estimated in a non-trivial number of industries (Panel A). The elasticity of plant-level revenues with respect to industry-revenues ($\hat{\beta}_q$) is on average a small positive number, but varies a lot across industries. Consequently, $\hat{\rho}$ and the demand elasticity ($\hat{\varepsilon}_{P_i}^{Q_i}$) and the markup ($1/\hat{\rho}$), also exhibit large variation. $\hat{\alpha}_j$-s are greater than $\hat{\beta}_j$-s in the average industry (see Panels B and C), which is to be expected under isoelastic demand. However, $\text{sd}(\hat{\alpha}_j) > \text{sd}(\hat{\beta}_j)$. Overall, these patterns suggests that although the variation in factor elasticities reflects $\text{var}(\hat{\beta}_j)$ and $\text{var}(\hat{\rho})$, the contribution of the latter likely dominates.

Table A12: Descriptive statistics of productivity distributions. Estimators are described in table 1, pre-estimation outliers included.

	N (1000)	IQR	SD	N (1000)	IQR	SD
	50 largest industries			10 largest industries		
OLS	568	0.29	0.28	230	0.23	0.21
OP	567	0.35	0.41	233	0.34	0.41
LPGR	572	0.33	0.35	235	0.32	0.31
WLPE	575	0.37	0.41	236	0.34	0.30
WLPM	575	0.46	2.21	236	0.37	0.39
GA	561	0.25	0.24	231	0.25	0.22

Table A13: Correlations among within-industry tfp-distributions, 50 largest 4-digit industries. Pre-estimation outliers included.

	OLS	OP	LPGR	WLPE	WLPM	GA
Pearson						
OLS	1					
OP	0.52	1				
LPGR	0.8	0.42	1			
WLPE	0.49	0.47	0.5	1		
WLPM	0.01	0.02	-0.15	0.15	1	
GA	0.77	0.47	0.63	0.49	0.09	
Spearman						
OLS	1					
OP	0.67	1				
LPGR	0.86	0.6	1			
WLPE	0.58	0.61	0.55	1		
WLPM	0.34	0.36	0.25	0.48	1	
GA	0.8	0.62	0.67	0.58	0.42	1

This table differs from table 4 in that we computed correlations including pre-estimation outlier observations.

Table A14: Correlations among within-industry tfp-distributions, 10 largest 4-digit industries.

	OLS	OP	LPGR	WLPE	WLPM	GA
Pearson						
OLS	1					
OP	0.48	1				
LPGR	0.71	0.43	1			
WLPE	0.71	0.54	0.8	1		
WLPM	0.52	0.27	0.41	0.66	1	
GA	0.88	0.42	0.68	0.66	0.5	1
Spearman						
OP	0.71	1				
LPGR	0.71	0.65	1			
WLPE	0.71	0.73	0.79	1		
WLPM	0.6	0.49	0.51	0.69	1	
GA	0.88	0.63	0.68	0.67	0.56	1

Including pre-estimation outliers barely changes correlations.

Table A15: Correlations among within-industry tfp-distributions, 10 largest 4-digit industries. Pre-estimation outliers included.

	OLS	OP	LPGR	WLPE	WLPM	GA
Pearson						
OLS	1					
OP	0.51	1				
LPGR	0.69	0.43	1			
WLPE	0.69	0.57	0.78	1		
WLPM	0.5	0.27	0.37	0.64	1	
GA	0.85	0.43	0.64	0.63	0.48	1
Spearman						
OLS	1					
OP	0.7	1				
LPGR	0.71	0.63	1			
WLPE	0.7	0.73	0.78	1		
WLPM	0.58	0.48	0.47	0.67	1	
GA	0.86	0.61	0.66	0.65	0.54	1

This table differs from table A14 in that we computed correlations including pre-estimation outlier observations.

Table A16: Monte Carlo results on productivity dispersion. Dispersion is measured using the interquartile range. Standard errors of iqr measures are bootstrapped and are in parentheses.

Panel 1. Pooled data

	orig. smpl	R=50	R=100
OLS	0.22	0.22	0.22
		(0.0004)	(0.0004)
OP	0.32	0.78	0.68
		(0.1014)	(0.0933)
LPGR	0.29	0.60	0.65
		(0.0706)	(0.0691)
WLPE	0.34	0.75	0.75
		(0.0794)	(0.0817)
WLPM	0.42	1.82	1.80
		(0.3788)	(0.3449)
GA	0.23	0.23	0.23
		(0.001)	(0.001)

Panel 2. Unweighted cross-industry average

	orig. smpl	R=50	R=100
OLS	0.26	0.26	0.26
		(0.0009)	(0.0008)
OP	0.36	0.39	0.39
		(0.0145)	(0.0203)
LPGR	0.36	0.38	0.38
		(0.0099)	(0.0111)
WLPE	0.58	1.02	1.05
		(0.4614)	(0.5588)
WLPM	1.37	1.94	1.95
		(1.1976)	(1.0269)
GA	0.26	0.26	0.26
		(0.0011)	(0.0011)

The moments of dispersion statistics in Panel 1 are calculated pooling data from all industries. Mean dispersion calculated in this manner can be interpreted as a weighted average across industries where the weights depend on the number of plant-year observations in an industry. This procedure is consistent with the way we calculated dispersion statistics in table 3. In Panel 2, dispersion is calculated as an unweighted cross-industry average of within-industry dispersion measures. Interestingly if we bootstrap dispersion statistics industry-by-industry and then calculate average dispersion (Panel 2), simulated results under proxy methods are closer to those in the original sample. Our MC approach implies that in this setup, larger and more heterogenous industries have smaller influence. The only exceptions are WLPM and WLPE, which are more likely to yield extreme elasticity estimates (see section 4) and therefore dispersion. For GA, coefficients are held constant while input and output observations vary across bootstrap replications. This exercise helps us assess the effect of the variation in input and output observations. It is natural to use GA coefficients for this exercise because the cost-shares are calculated using industry level information.

Table A17: Sample size in the specifications shown in table 5. Sample size is measured as the total number of plant-year observations used in a regression (in thousands).

	OLS	OP	LPGR	WLPE	WLPM	GA
4-digit elasticities						
$dlnE$	410	415	413	414	414	405
exit	410	415	413	414	414	405
$dlnE_{cont}$	393	398	396	397	397	388
3-digit elasticities						
$dlnE$	405	413	414	413	413	405
exit	405	413	414	413	413	405
$dlnE_{cont}$	389	396	398	396	396	388
3-digit elasticities, industries with negative or non-estimable elasticities are dropped						
$dlnE$	391	406	408	289	310	405
exit	391	406	408	289	310	405
$dlnE_{cont}$	375	390	391	277	298	388
Total N	440	440	440	440	440	440

Total N denotes the original number of plant-year observations before estimation and post-estimation outlier trimming.

Table A18: The effect of productivity on outcomes, all (292) largest industries. Outcomes are: employment growth among all establishments, exit and employment growth among continuers

	OLS	OP	LPGR	WLPE	WLPM	GA
4-digit elasticities						
$dlnE$	0.183***	0.001	0.128***	0.004**	-0.004***	0.203***
exit	-0.066***	0.001	-0.041***	-0.002*	0.001**	-0.068***
$dlnE_{cont}$	0.055***	0.002***	0.05***	0.001	-0.001*	0.071***
3-digit elasticities						
$dlnE$	0.202***	0.115***	0.15***	0.001	-0.012***	0.193***
exit	-0.072***	-0.037***	-0.045***	0.000	0.004**	-0.066***
$dlnE_{cont}$	0.062***	0.046***	0.067***	0.000	-0.005***	0.066***
3-digit elasticities, industries with negative or non-estimable elasticities are dropped						
$dlnE$	0.207***	0.159***	0.169***	0.078***	0.06***	0.193***
exit	-0.073***	-0.05***	-0.05***	-0.031***	-0.022***	-0.066***
$dlnE_{cont}$	0.065***	0.065***	0.075***	0.017***	0.017***	0.066***

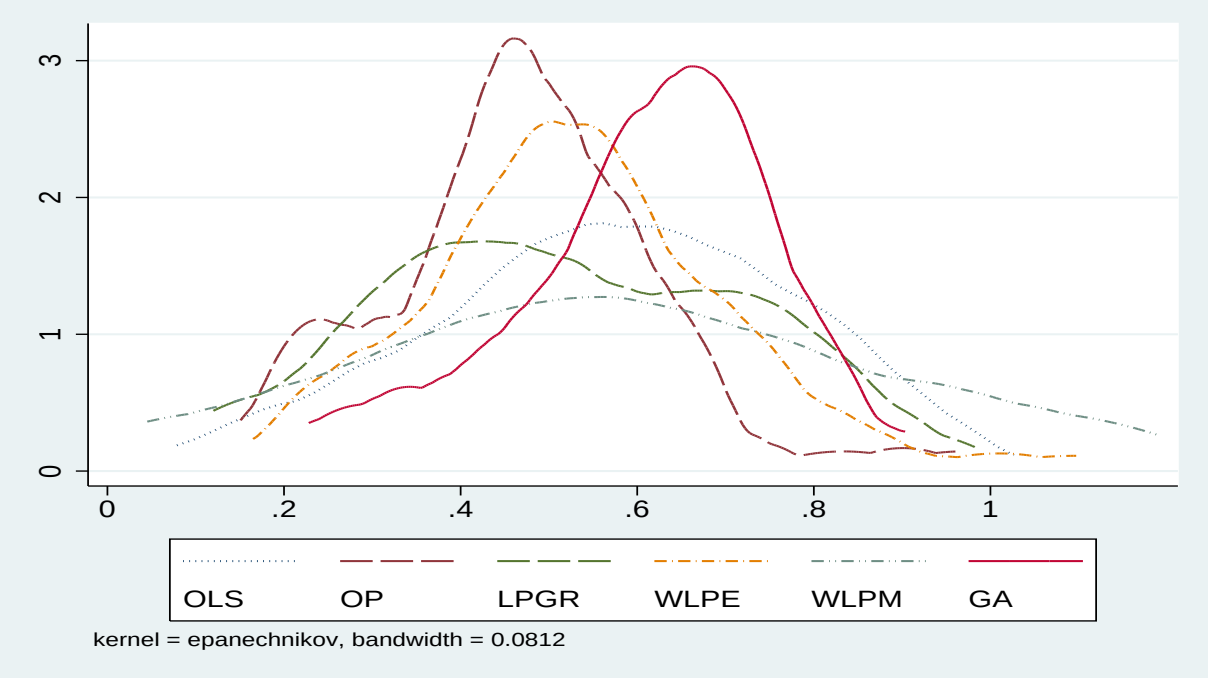
Estimates are taken from equations of three outcomes on plant-level measure of productivity, state-level measure of unemployment growth, and year-, sizeclass- and state-fixed effects. Standard errors are clustered at the state level, asterisks denote standard significance levels.

Table A19: The effect of productivity on outcomes, 10 largest industries. Outcomes are: employment growth among all establishments, exit and employment growth among continuers

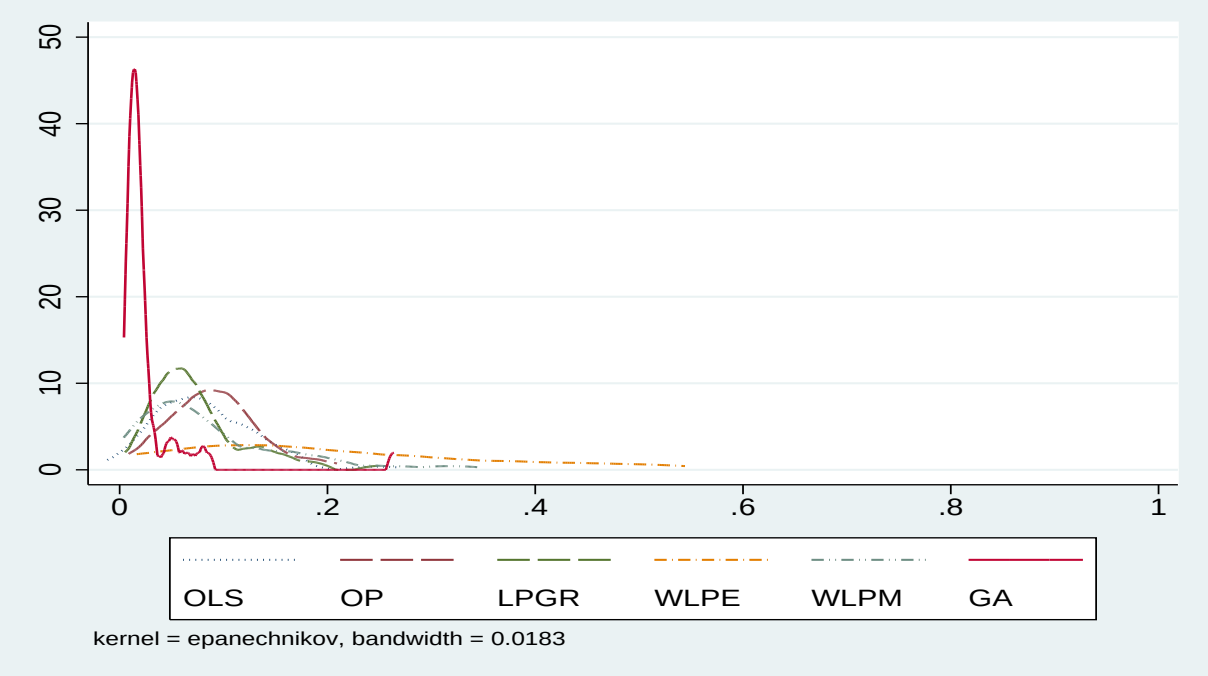
	OLS	OP	LPGR	WLPE	WLPM
4-digit elasticities					
$dlnE$	0.206***	0.051***	0.154***	0.099***	-0.027**
exit	-0.067***	-0.002	-0.046***	-0.022***	0.014***
$dlnE_{cont}$	0.076***	0.054***	0.066***	0.06***	0.003
3-digit elasticities					
$dlnE$	0.208***	0.158***	0.164***	0.109***	0.016
exit	-0.068***	-0.032***	-0.049***	-0.026***	0.002
$dlnE_{cont}$	0.075***	0.102***	0.07***	0.061***	0.023*
3-digit elasticities, industries with negative or non-estimable elasticities are dropped					
$dlnE$	0.208***	0.158***	0.164***	0.13***	0.061***
exit	-0.068***	-0.032***	-0.049***	-0.043***	-0.018***
$dlnE_{cont}$	0.075***	0.102***	0.07***	0.046***	0.027*

Estimates are taken from equations of three outcomes on plant-level measure of productivity, state-level measure of unemployment growth, and year-, sizeclass- and state-fixed effects. Standard errors are clustered at the state level. *, ** and *** denote 10%, 5% and 1% significance levels.

A.7 Additional Figures



(a) $\hat{\beta}_m$



(b) $\hat{\beta}_e$

Figure A3: Cross-industry distributions of elasticities of $\hat{\beta}_m$ and $\hat{\beta}_e$.
The sample is 50 largest industries, estimators are described in table 1. $\hat{\beta}_m$-s show similar general shape but differences exist. For $\hat{\beta}_e$, results are also more similar except GA and WLPE.

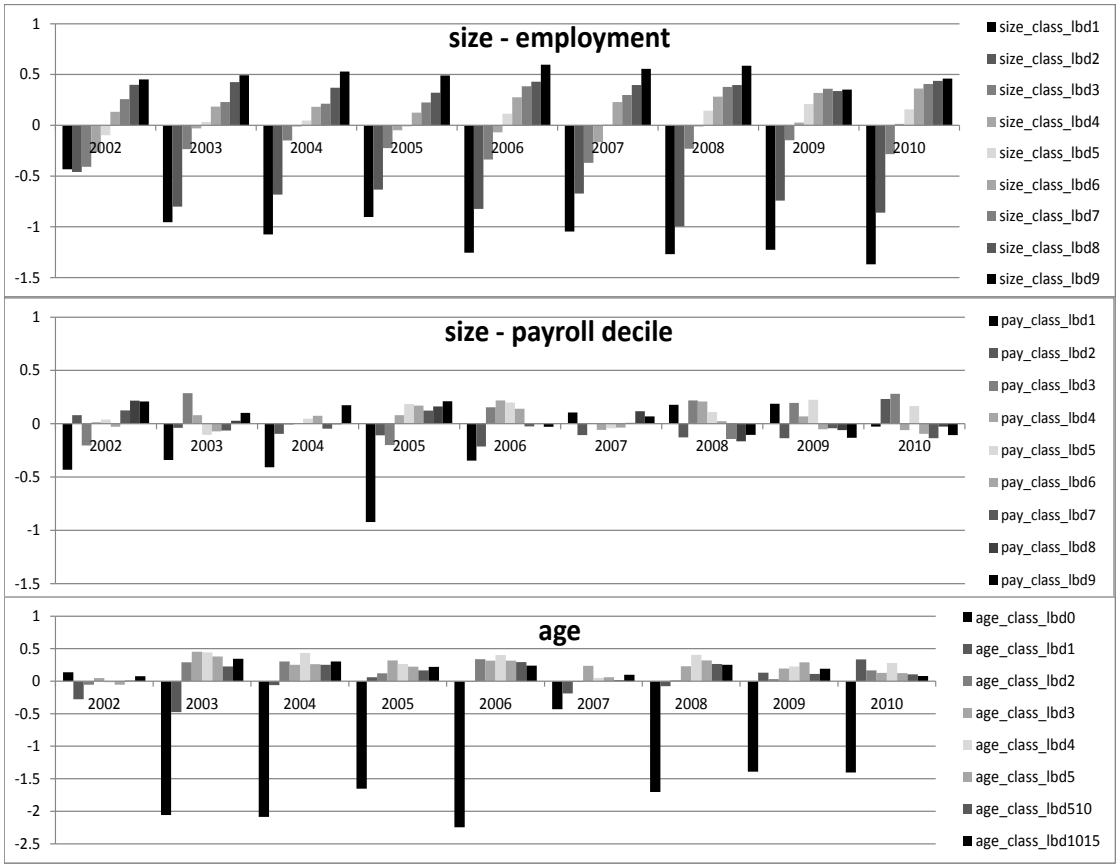


Figure A4: Point estimates from a logistic regression of the probability of productivity being non-imputed.

Regressors: industry fixed effects, employment-, payroll-size and age classes. ASM cases only. We defined 10 size classes based on employment (1-9, 10-109, 20-29, 30-49, 50-99, 100-149 150-249, 250-499, 500-999, 1000+) in addition to the payroll deciles. Finally, we classified establishments into 9 age classes (births, 1, 2, 3, 4, 5, 6-10, 11-15, 16+ years).